INCIDENT ANALYSIS REPORT

Task # 03: Malware Download and Incident Response Plan

- **Submitted By: Raja Muhammad Bilal Arshad**
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- **Organization: ITSOLERA (PVT) LTD**

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EXECUTIVE OVERVIEW

This report provides an objective analysis of a simulated malware infection and network intrusion. It details the end-to-end detection, logging, and comprehensive incident response strategy executed across both endpoint and network perimeters. The simulation encompasses the deployment of custom Network Intrusion Detection Systems (NIDS) , advanced endpoint telemetry , and centralized SIEM aggregation to detect both simulated Command and Control (C2) beaconing and the execution of the EICAR standard anti-virus test file.

TARGET ENVIRONMENT & AGENTS

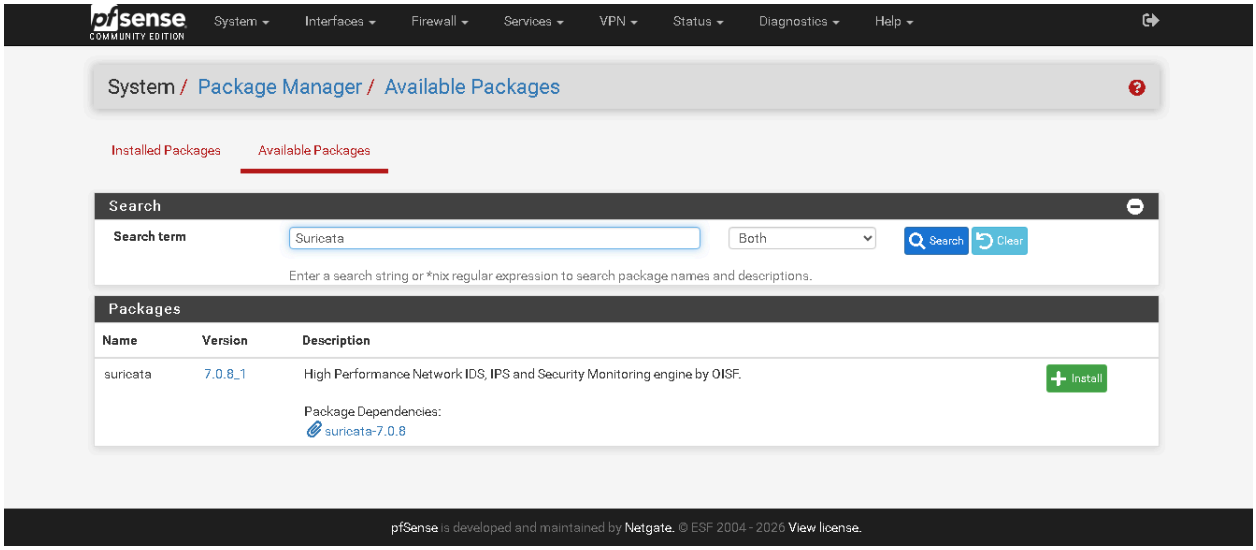
1. **Target Endpoint:** Windows 10 Virtual Machine
2. **Endpoint Hostname:** DESKTOP-4VFDF67
3. **Wazuh Agent Name:** Joker
4. **Compromised Host IP:** 192.168.1.100
5. **Web Server / Attacker Host:** Nginx

SECURITY STACK & TOOLS DEPLOYED

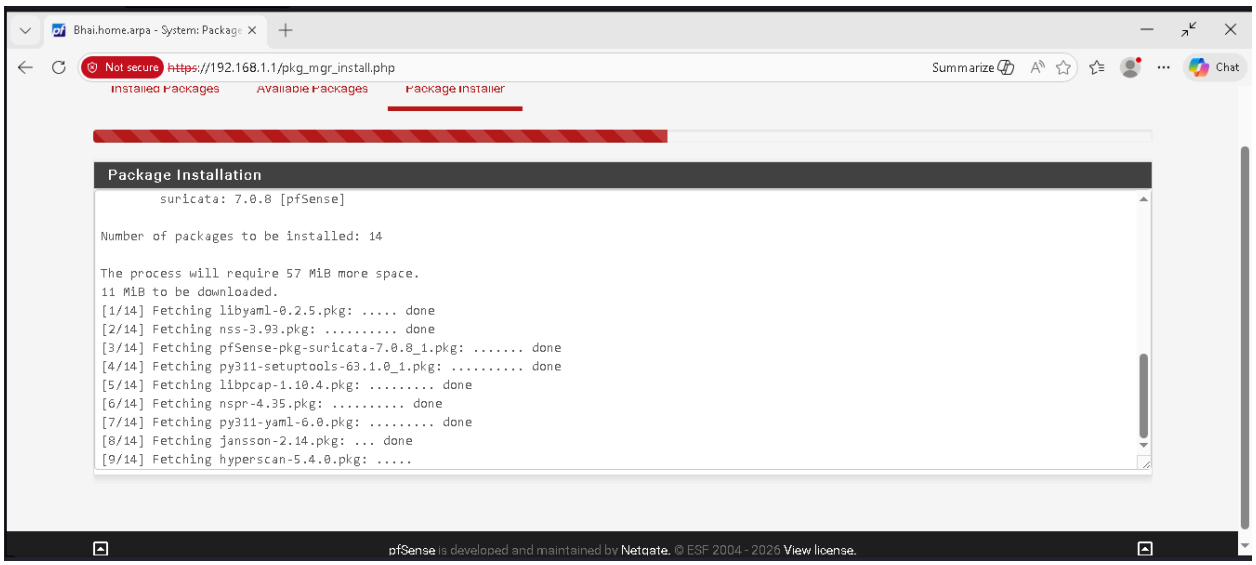
1. **SIEM & Log Aggregation:** Wazuh Manager
2. **Endpoint Telemetry & Behavioral Monitoring:** Microsoft Sysinternals Sysmon (v15.15) paired with Olaf Hartong's Sysmon-Modular Configuration
3. **Perimeter Firewall & Routing:** pfSense
4. **Network Intrusion Detection Systems (NIDS):** Suricata and Snort (Deployed via pfSense Package Manager)
5. **Endpoint Protection Platform (EPP):** Microsoft Defender Antivirus
6. **Log Forwarding & Packet Analysis:** syslog-ng , tcpdump
7. **Command Line & Execution Tools:** Windows PowerShell , cURL
8. **Configuration Editors:** Pluma
9. **Simulated Malware & Payloads:** * EICAR Standard Anti-Virus Test File
    a. Simulated C2 Beaconing via HTTP GET request (User-Agent spoofed as "BlackSun")

Phase 1: Network Intrusion Detection System (NIDS) Setup on pfSense

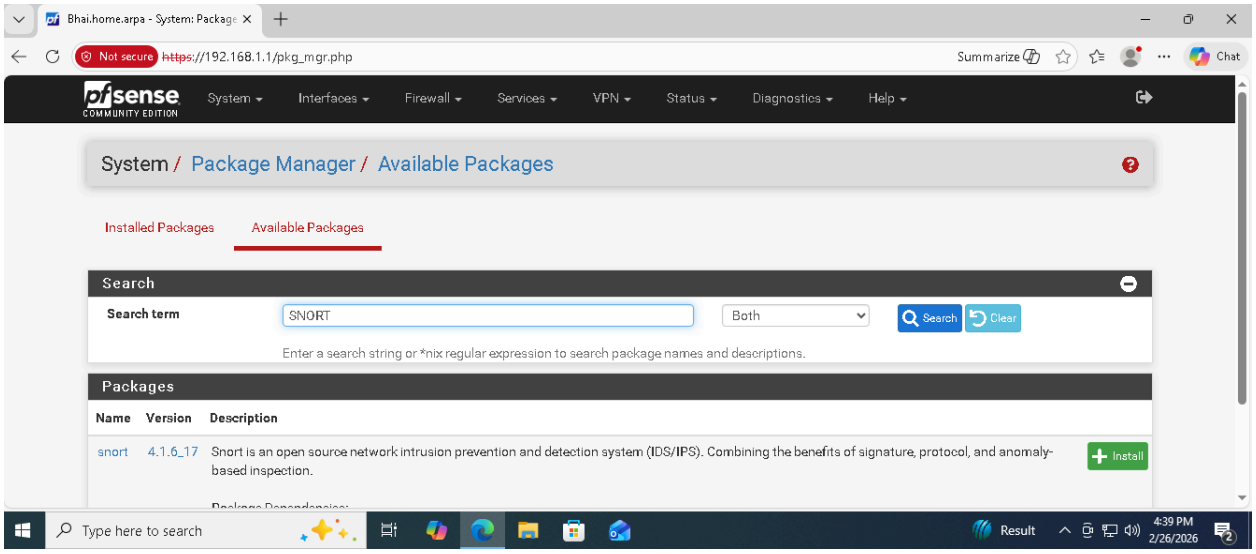
Description for Suricata Package Installation: To actively monitor and analyze network traffic for malicious signatures, the Suricata package was selected for deployment on the pfSense firewall. This screenshot demonstrates the pfSense Package Manager interface, showing Suricata available for installation to provide high-performance Network Intrusion Prevention System (IPS) and Intrusion Detection System (IDS) capabilities.



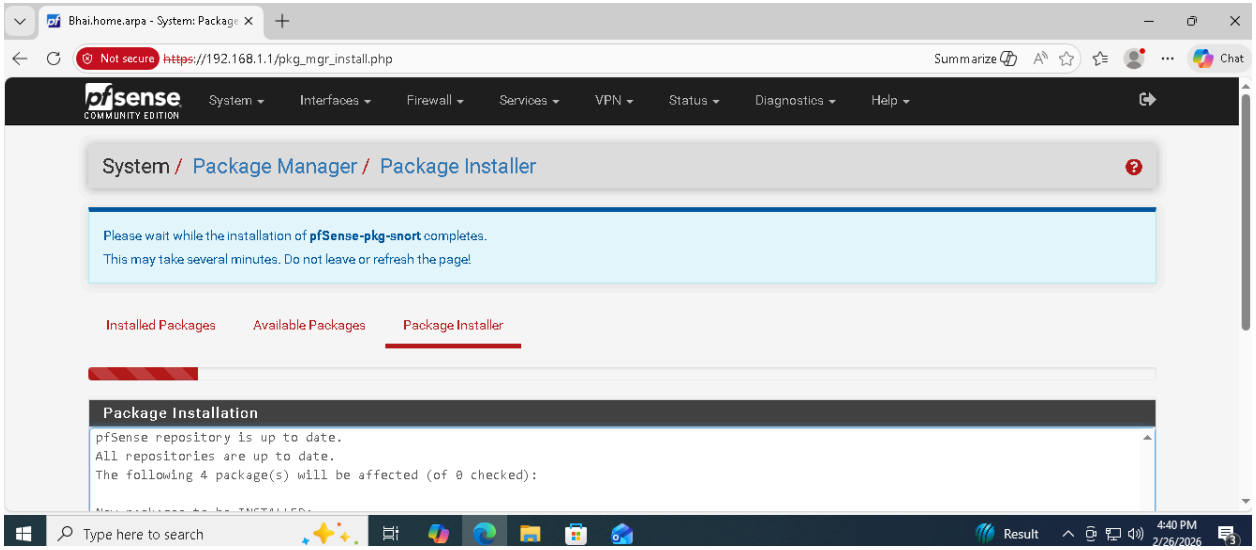
Description for Suricata Package Extraction & Dependencies: This image captures the pfSense Package Installer actively fetching and installing the necessary dependencies and libraries required for Suricata to operate successfully. It confirms the successful retrieval of packages like libpcap and hyperscan, ensuring the engine can perform deep packet inspection.



Description for Snort Package Installation:
As an alternative or supplementary detection engine, Snort was also searched within the pfSense Package Manager. This screenshot illustrates the availability of the Snort open-source network intrusion prevention and detection system package for deployment.



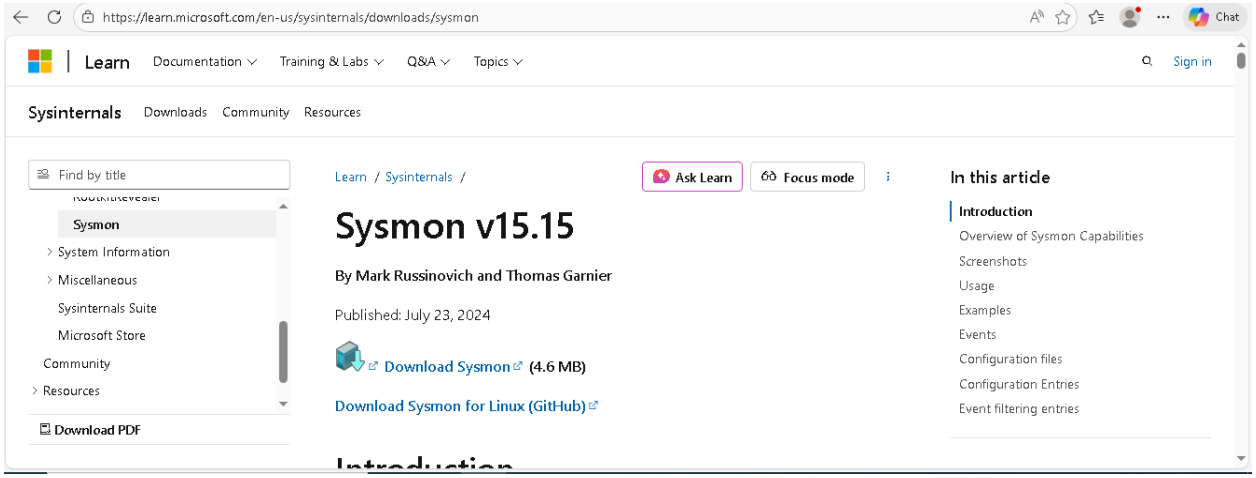
Description for Snort Package Extraction: Similar to Suricata, this screenshot displays the automated package installation process for Snort, verifying that the pfSense repository is up to date and that the required packages are being fetched without error.



Phase 2: Endpoint Agent Configuration (Sysmon Setup)

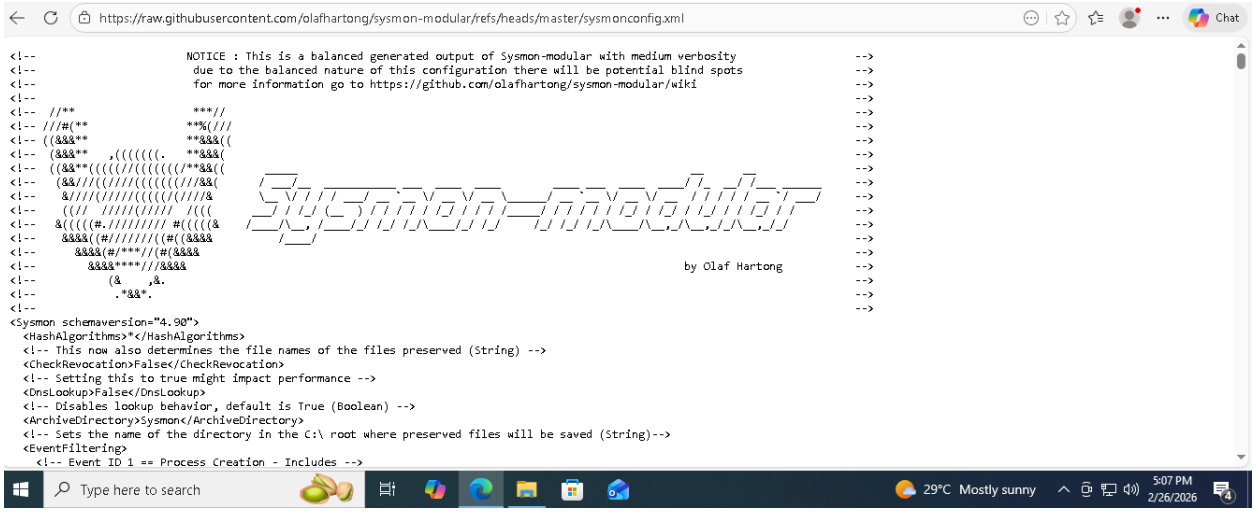
Description for Sysmon Download:

To achieve granular visibility into endpoint activities, such as process creation, network connections, and file modifications, Microsoft Sysinternals Sysmon v15.15 was downloaded. This screenshot shows the official Microsoft documentation and download page used to acquire the binary for the Windows 10 target endpoint.

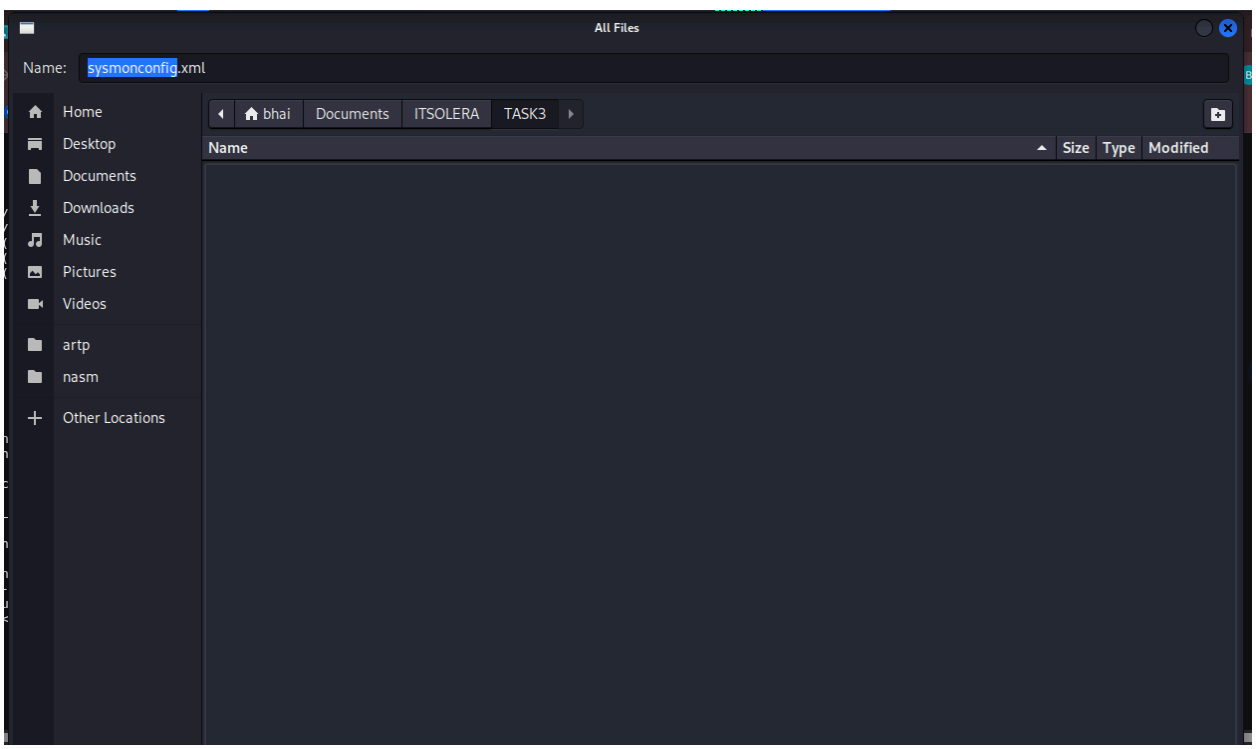


Description for Sysmon-Modular Configuration Acquisition: By default, Sysmon logs a massive amount of data. To filter out noise and focus on high-fidelity security events, a community-driven, balanced configuration file (sysmonconfig.xml) by Olaf Hartong was utilized. This screenshot shows the raw XML file hosted on GitHub, which maps events to the MITRE ATT&CK framework.

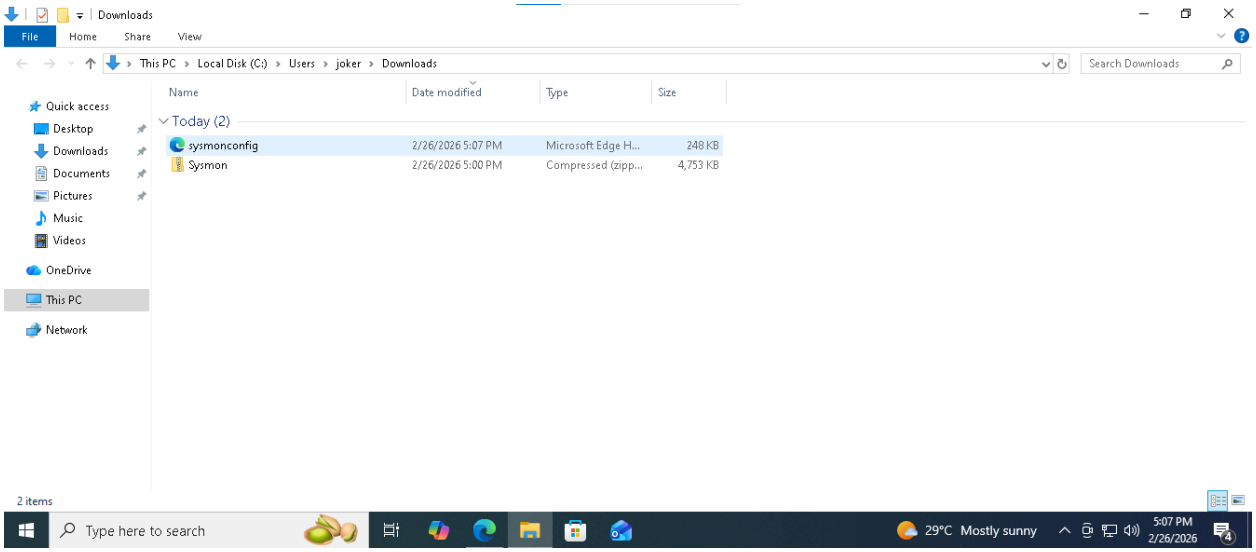
Open from the github sysmonconfigs.xml and open it as RAW



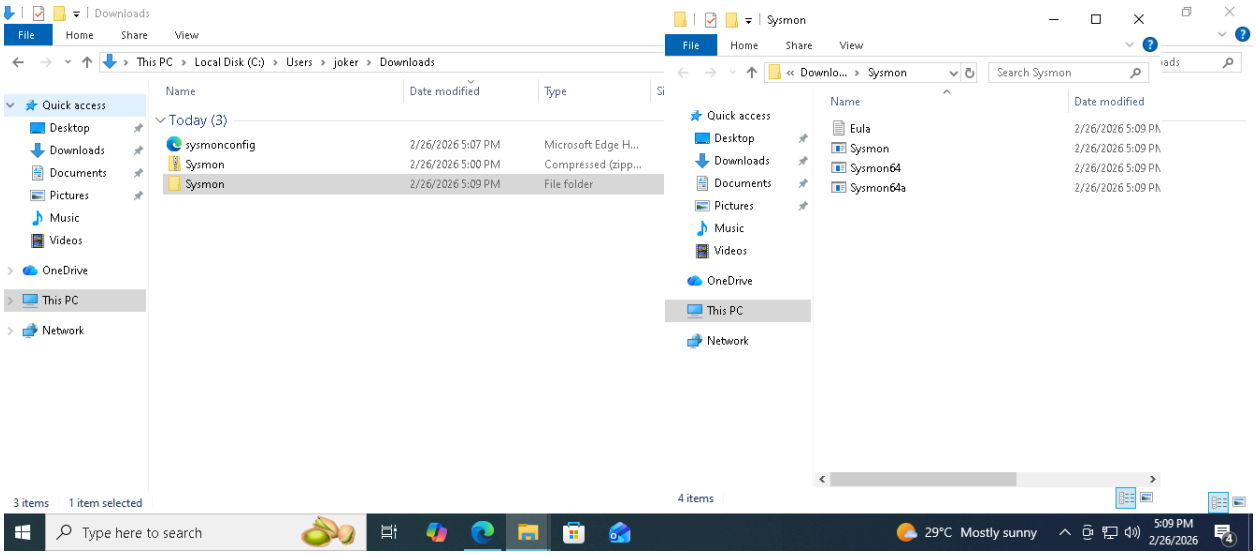
Description for Saving the Configuration File: This image shows the process of saving the raw sysmonconfig.xml file locally to the system so it can be passed as an argument during the Sysmon installation phase.



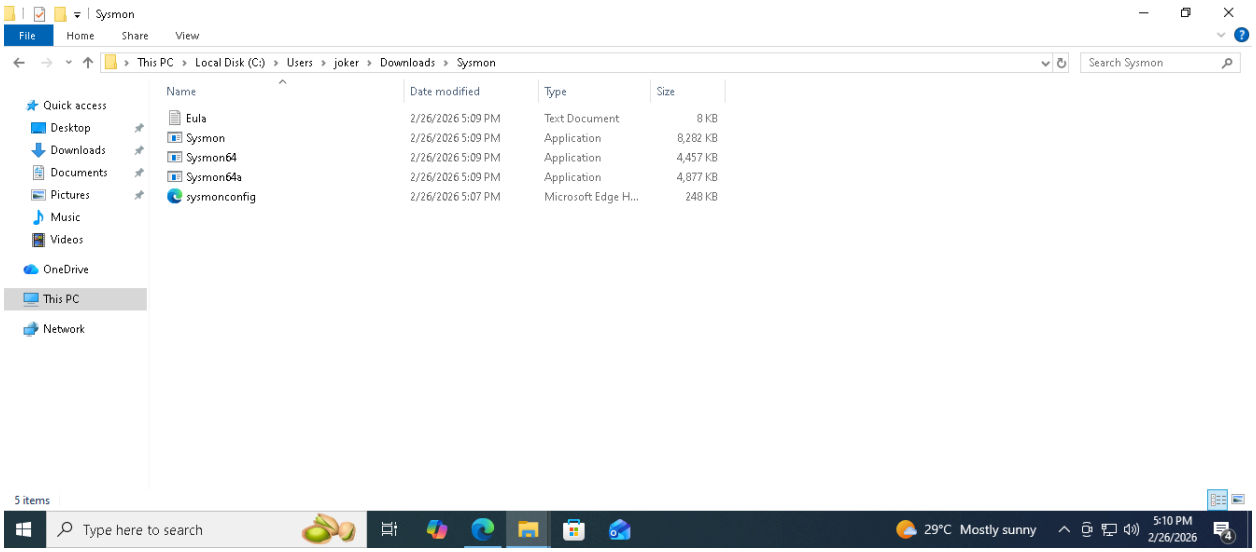
Description for Organizing Sysmon Binaries: To maintain an organized workspace for the installation, the downloaded Sysmon zip archive was extracted into a dedicated folder within the Downloads directory, as shown in this Windows File Explorer screenshot.



Description for Sysmon Extraction Verification: This screenshot confirms the successful extraction of the Sysmon utilities, displaying the executables (including Sysmon64.exe) and the EULA document.

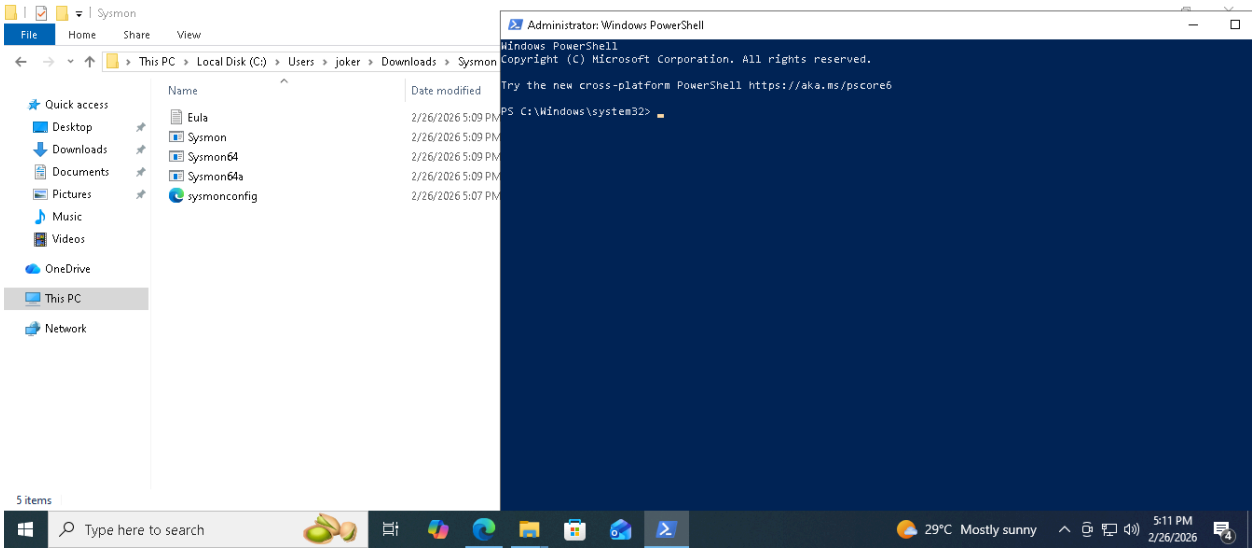


Description for Staging the Configuration File: For ease of execution via the command line, the previously downloaded sysmonconfig.xml was moved directly into the extracted Sysmon folder alongside the Sysmon64.exe binary.



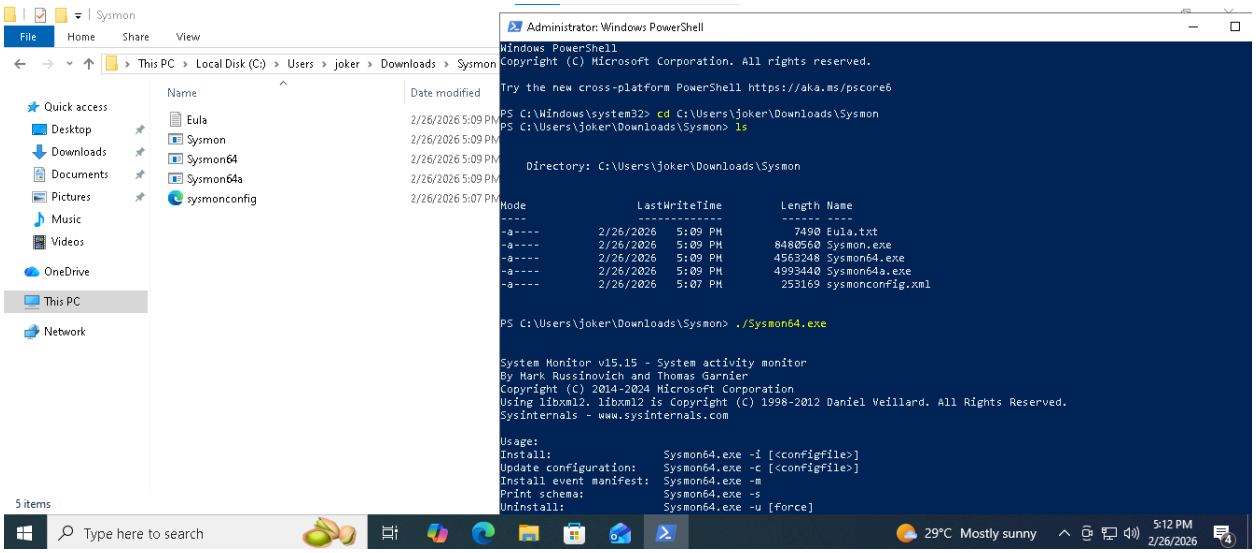
Phase 3: Sysmon Deployment and Verification

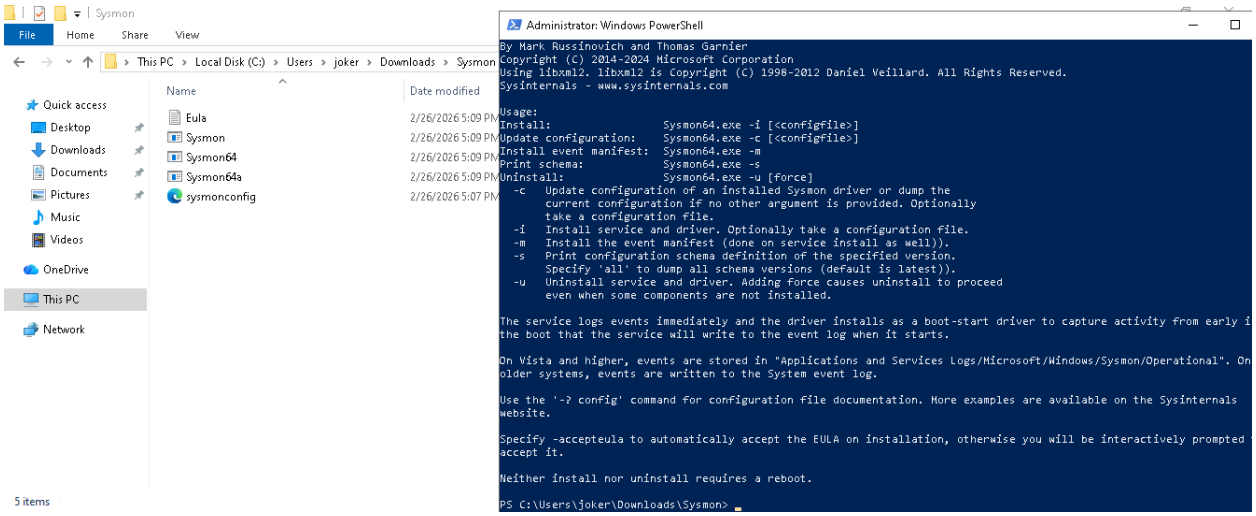
Description for Launching PowerShell as Administrator: To install a system-level driver and service, administrative privileges are required. This screenshot demonstrates opening an elevated Windows PowerShell session and navigating to the staged Sysmon directory.



Description for Reviewing Sysmon Help Menu: Before installation, the ./sysmon64.exe command was run without arguments to review the available flags and confirm the binary was executing correctly. The

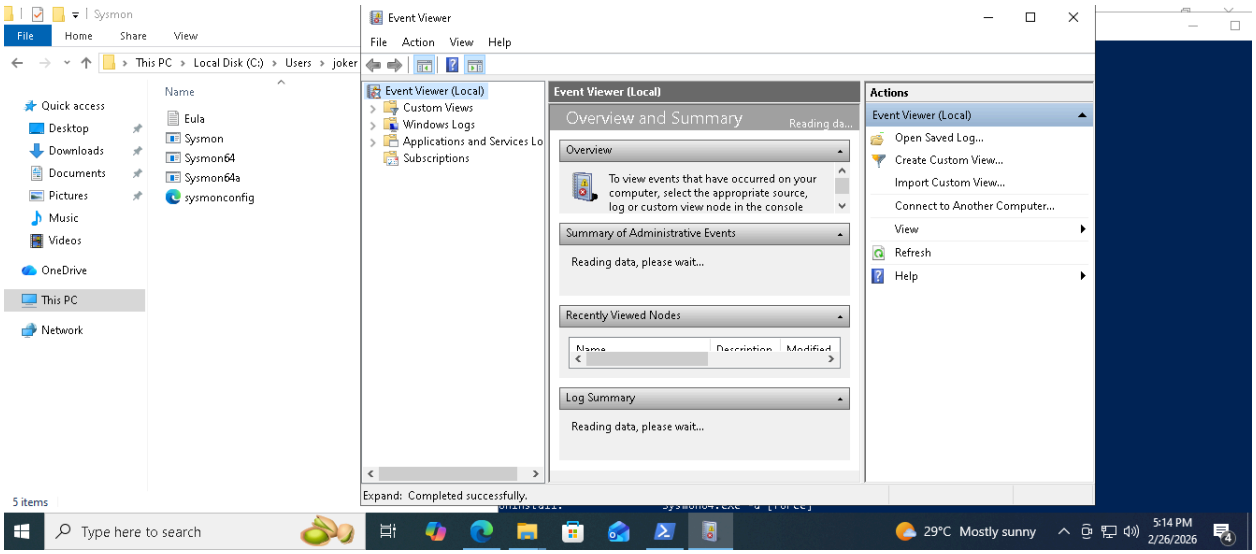
output details the commands for updating configurations and accepting the End User License Agreement (EULA).



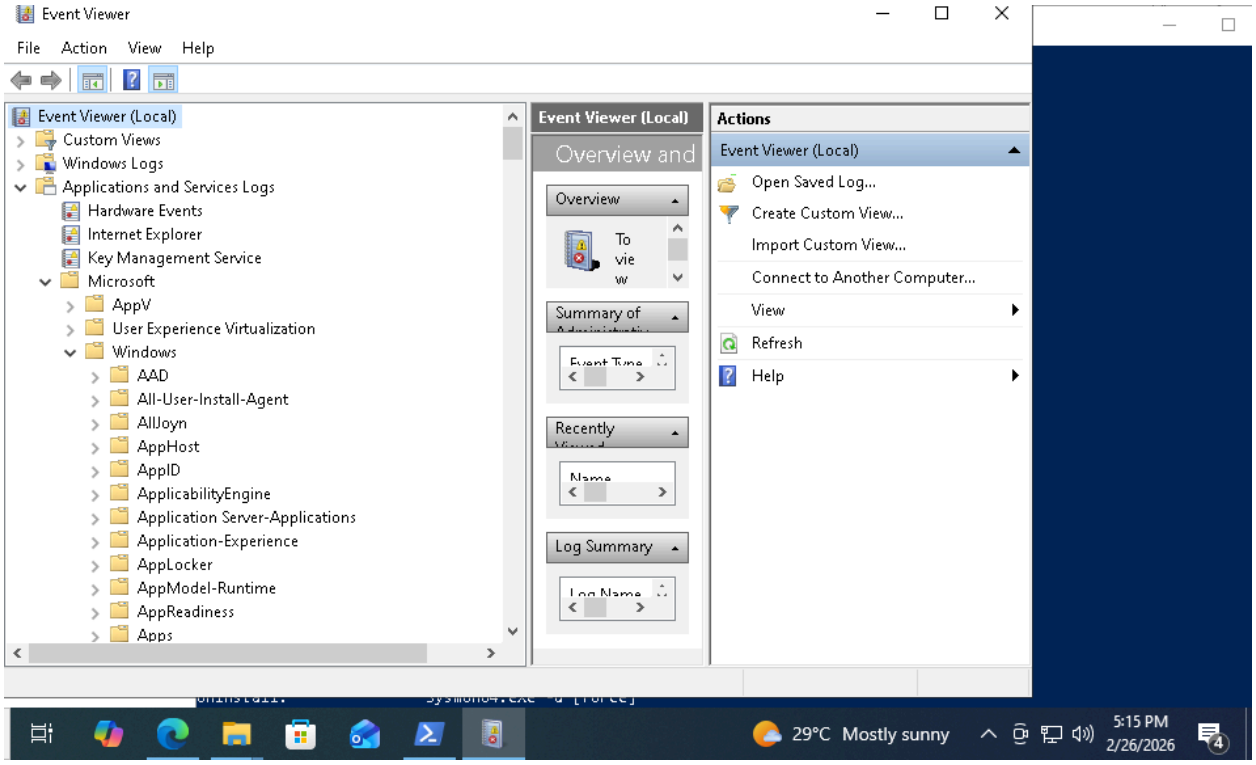


Now check if Sysmon installed in your system or not
Open The Event Viewer.

Description for Pre-Installation Verification via Event Viewer: Before installing Sysmon, it is critical to verify the current state of the system logs. This screenshot shows the Windows Event Viewer navigating to Applications and Services Logs -> Microsoft -> Windows.

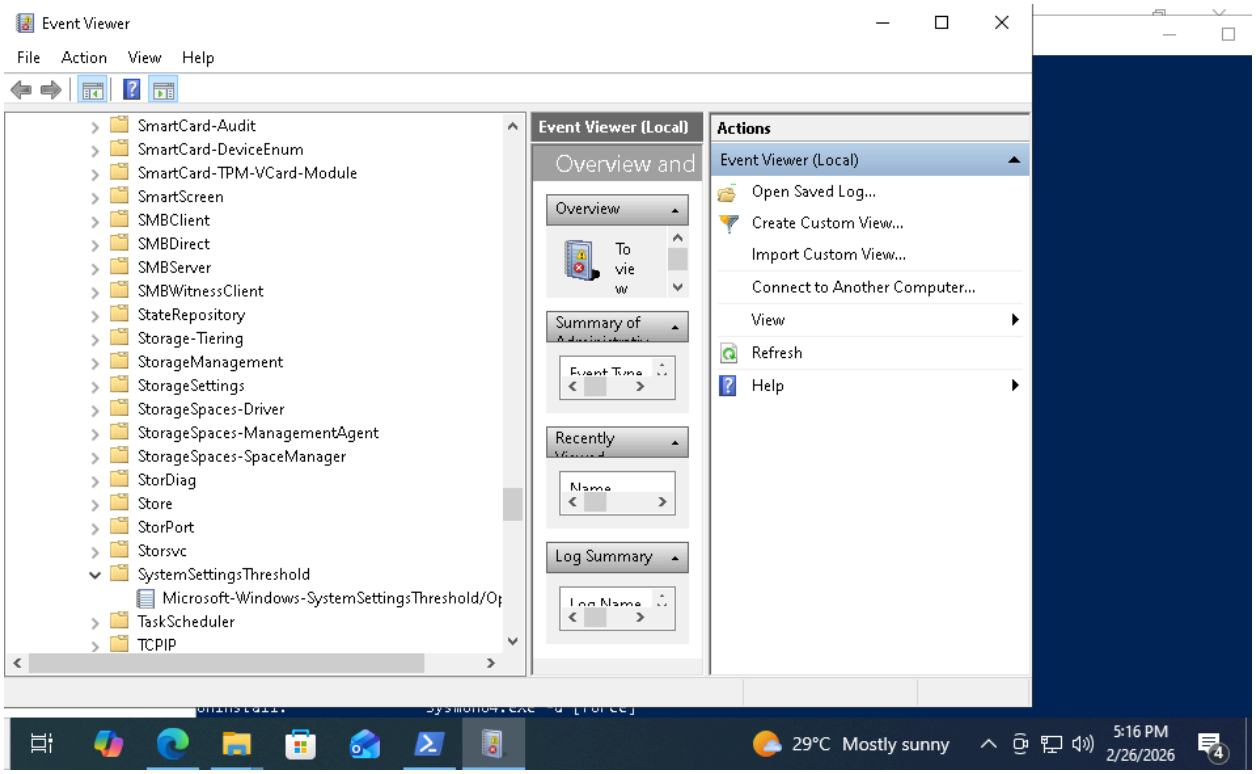


nOW fULLY OPEN THE WINDWOWS DROP DOWN MENU



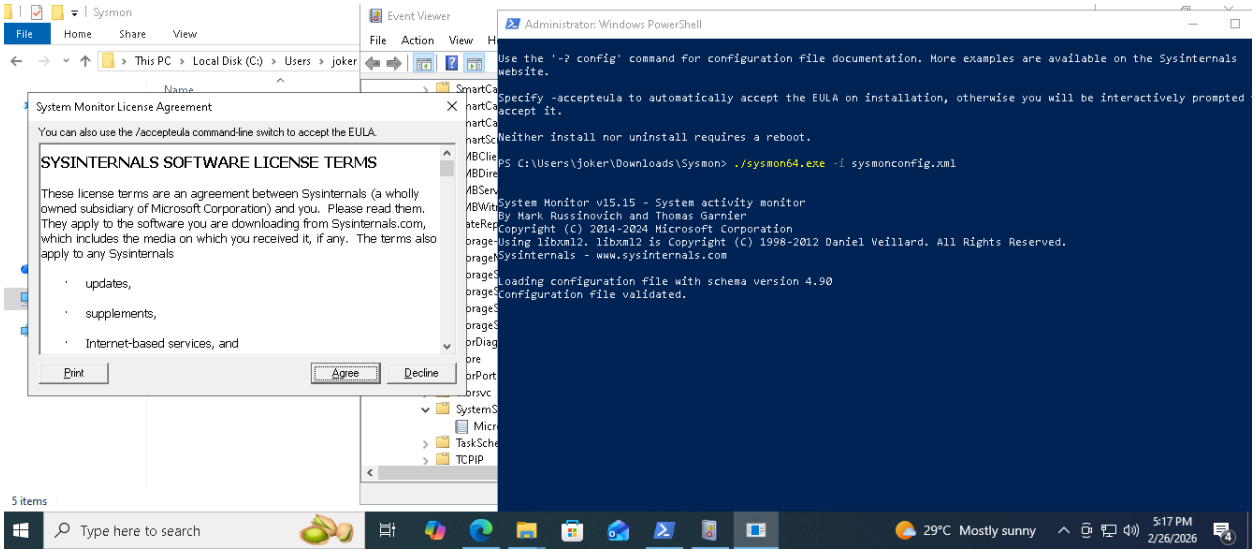
Now you will see no Sysmon here!!

Description for Confirming Absence of Sysmon Logs: Expanding the Microsoft Windows log directory confirms that the Sysmon folder does not yet exist, establishing a baseline and proving that the service is not currently installed on the target endpoint.



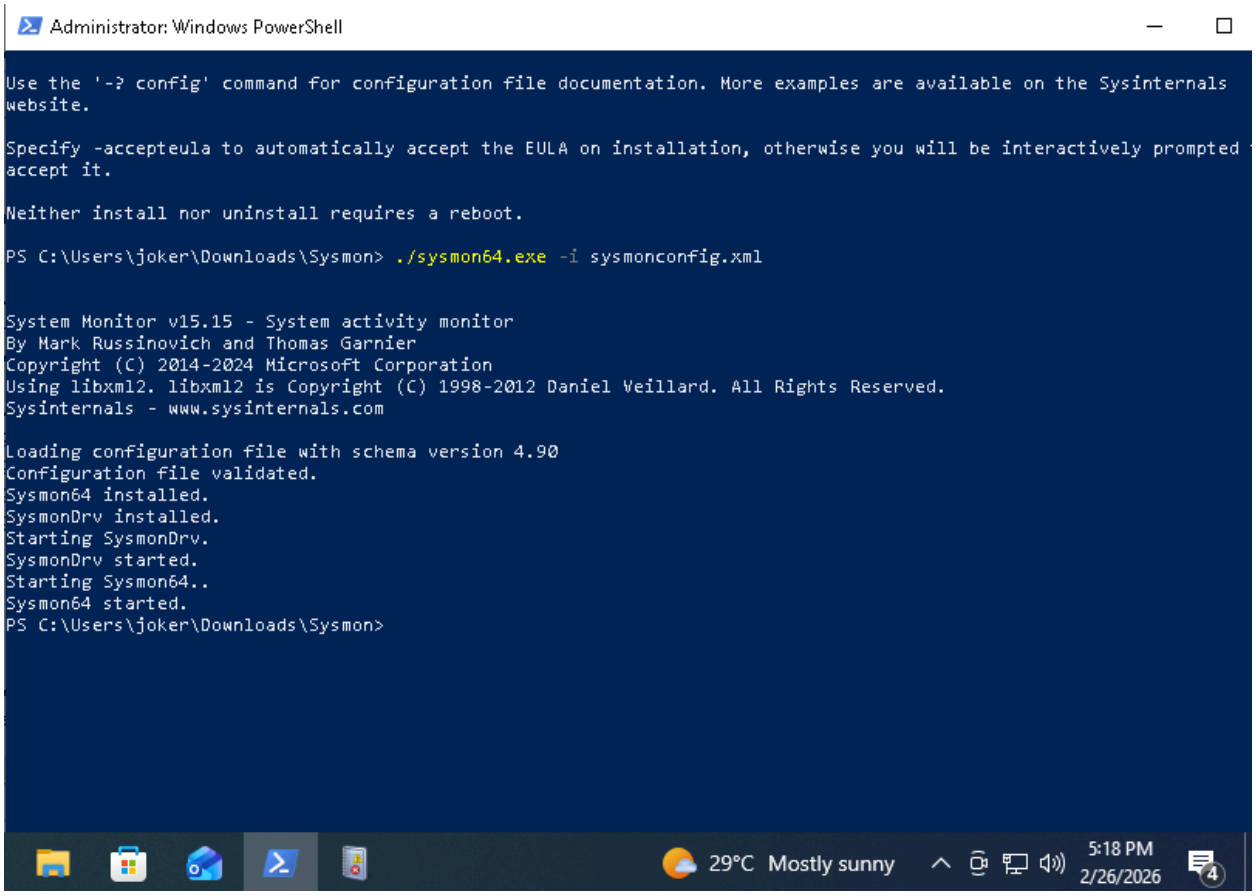
Then go back to the Powershell and type this :
"./sysmon64.exe -i sysmonconfig.xml" Which is work and open it

Description for Sysmon EULA Acceptance: Executing the installation command ./sysmon64.exe -i sysmonconfig.xml triggers the Sysinternals Software License Terms prompt. This GUI popup requires explicit agreement to install the system-level driver.



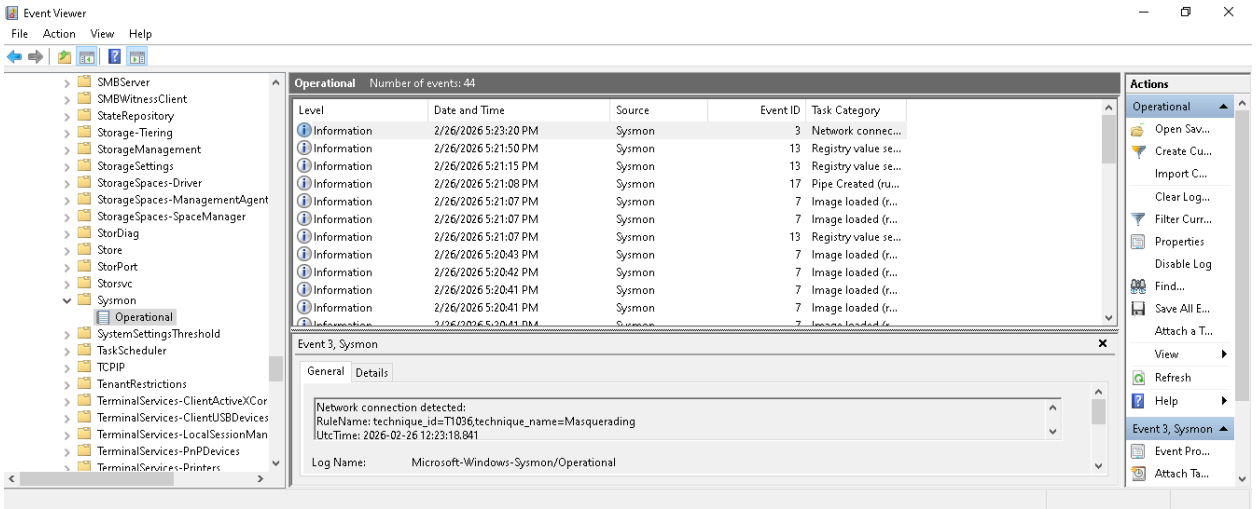
Click on Agree and it will install everything in it

Description for Successful Sysmon Installation: Following the EULA acceptance, PowerShell outputs confirmation that the configuration file schema (version 4.90) was validated, the Sysmon64 service was installed, and the SysmonDrv driver successfully started.



Now close the previous tab of the event Viewer and open a new event viewer and now you will find the sysmon in the event viewer:

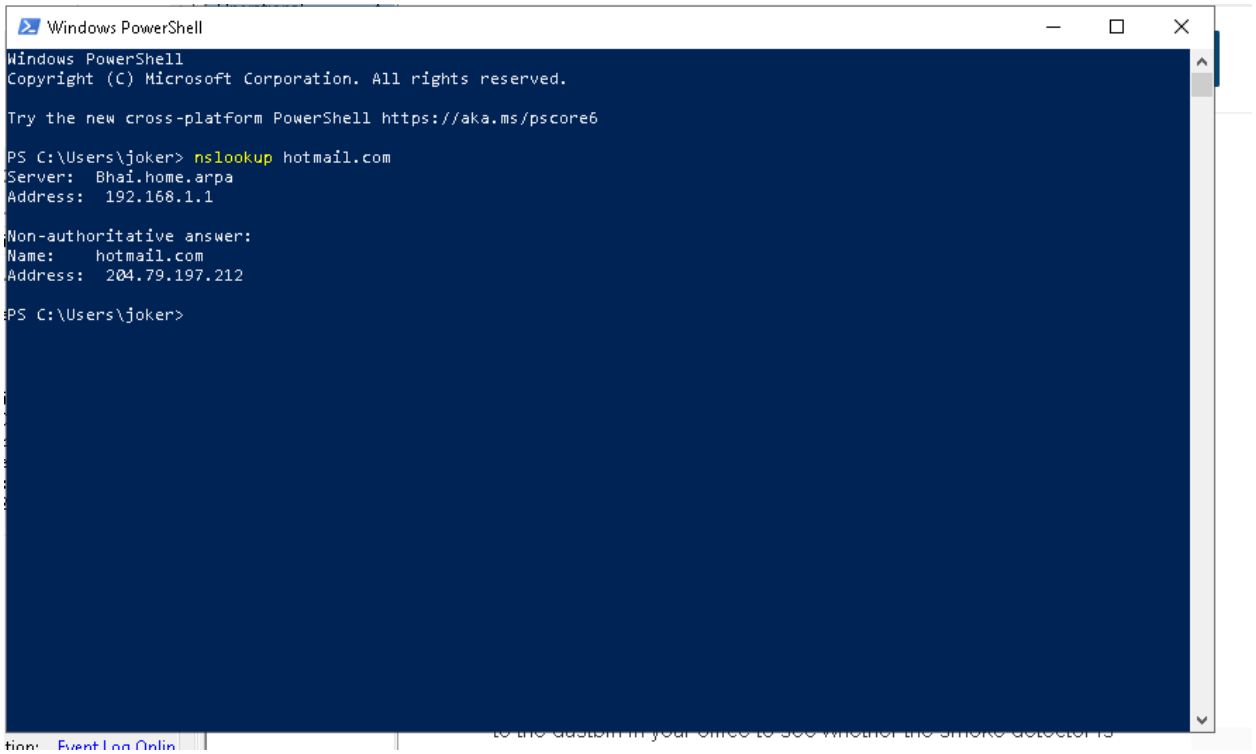
Description for Post-Installation Event Viewer Verification: With Sysmon successfully running, opening a fresh instance of the Event Viewer reveals the newly created Sysmon/Operational log channel. The screenshot shows immediate telemetry generation, including Event ID 7 (Image loaded) and Event ID 13 (Registry value set), proving the agent is actively monitoring the system.



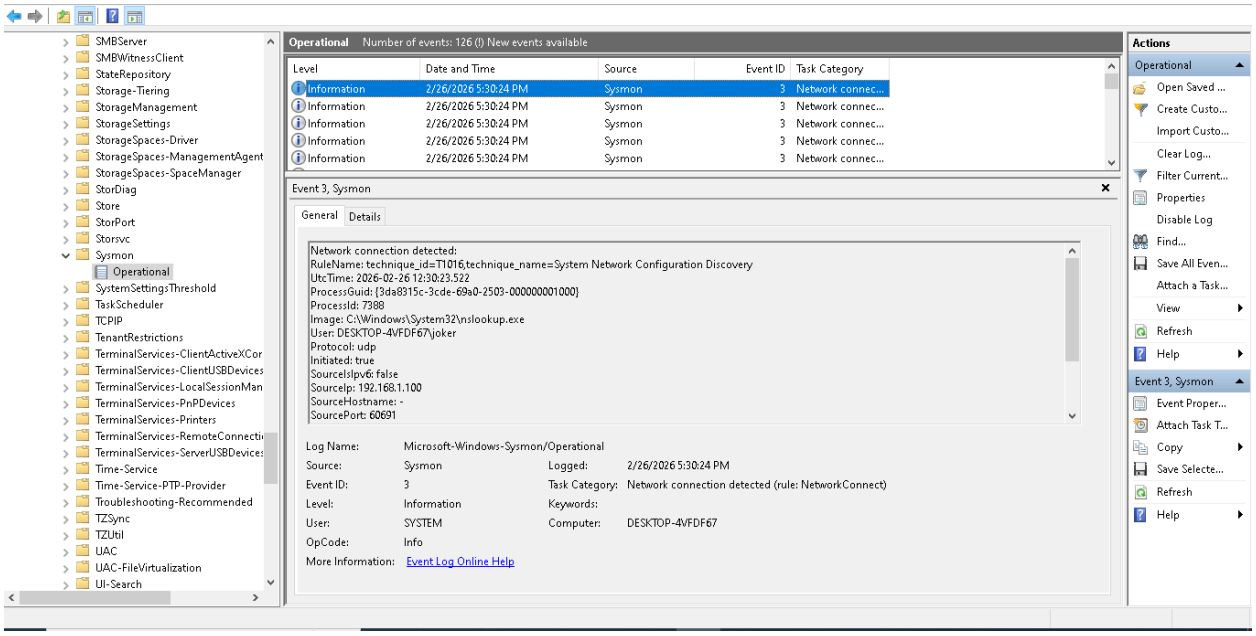
Now test everything:

Phase 4: Validating Endpoint and Network Telemetry

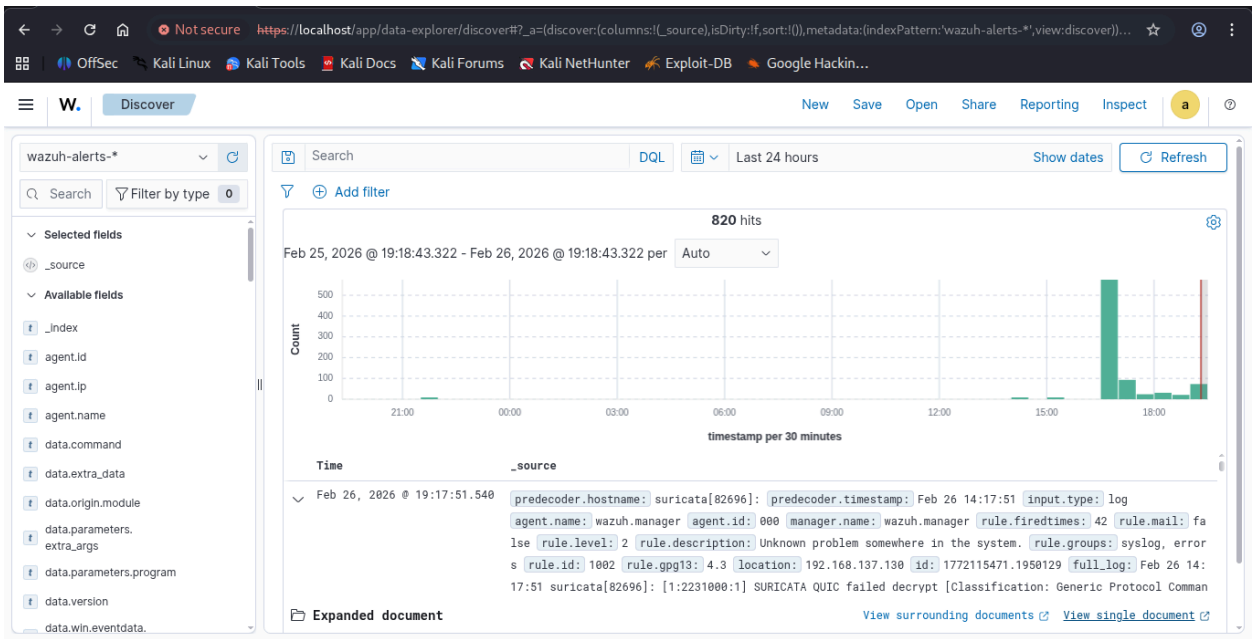
Description for Generating Network Telemetry (nslookup): To test Sysmon's network monitoring capabilities (Event ID 3), an nslookup hotmail.com command was executed in PowerShell. This forces the endpoint to make a DNS query, which should be immediately captured by the newly installed driver.



Description for Validating Sysmon Event ID 3 (Network Connection): Returning to the Event Viewer, we can see Sysmon successfully captured the DNS request. The log details show Event ID 3 (Network connection detected) originating from C:\Windows\System32\nslookup.exe (ProcessId: 7188) communicating over UDP port 53. Furthermore, the modular configuration correctly mapped this to MITRE ATT&CK technique T1016 (System Network Configuration Discovery).



Description for Testing the Intrusion Detection System (IDS): To ensure both the endpoint agent and the network firewall are communicating with the SIEM, a benign test alert was generated by curling a known NIDS testing URL: http://testmynids.org/uid/index.html. The HTTP 200 OK response confirms the payload crossed the network.



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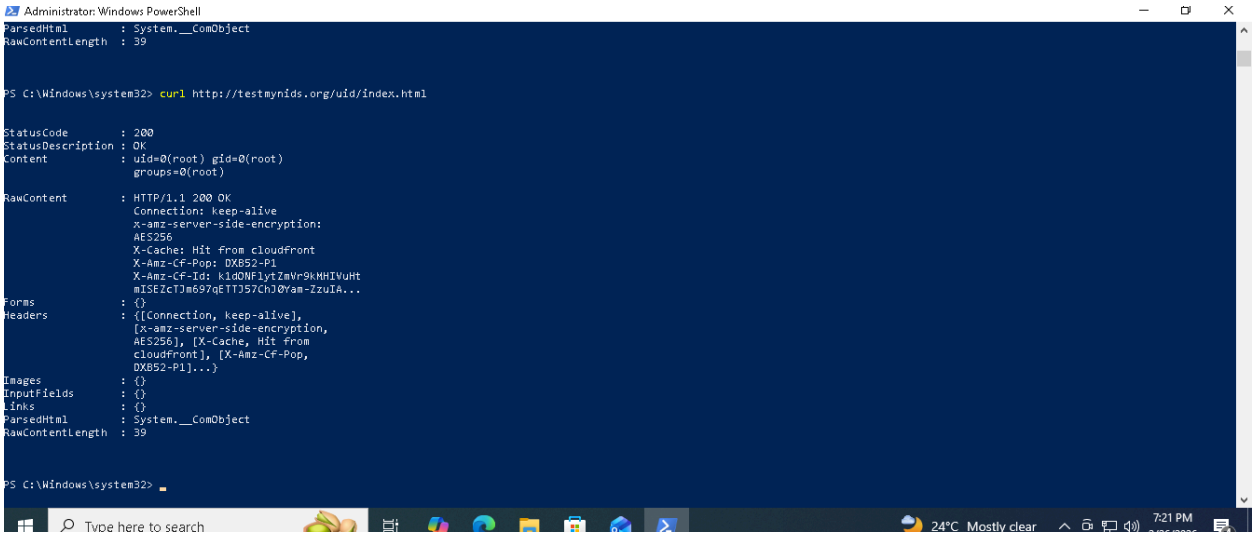
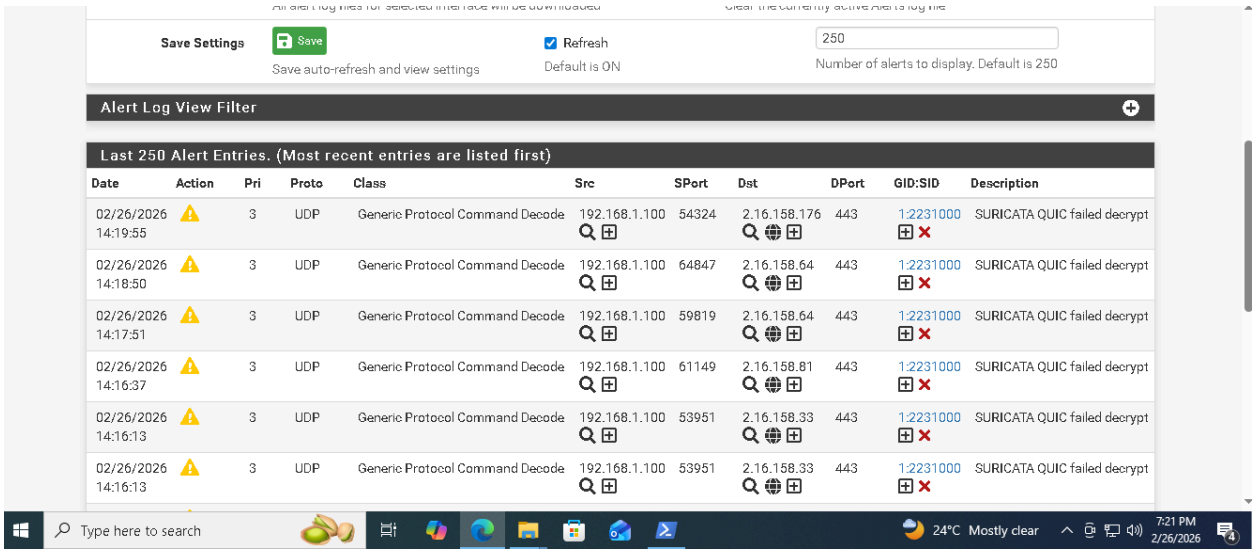
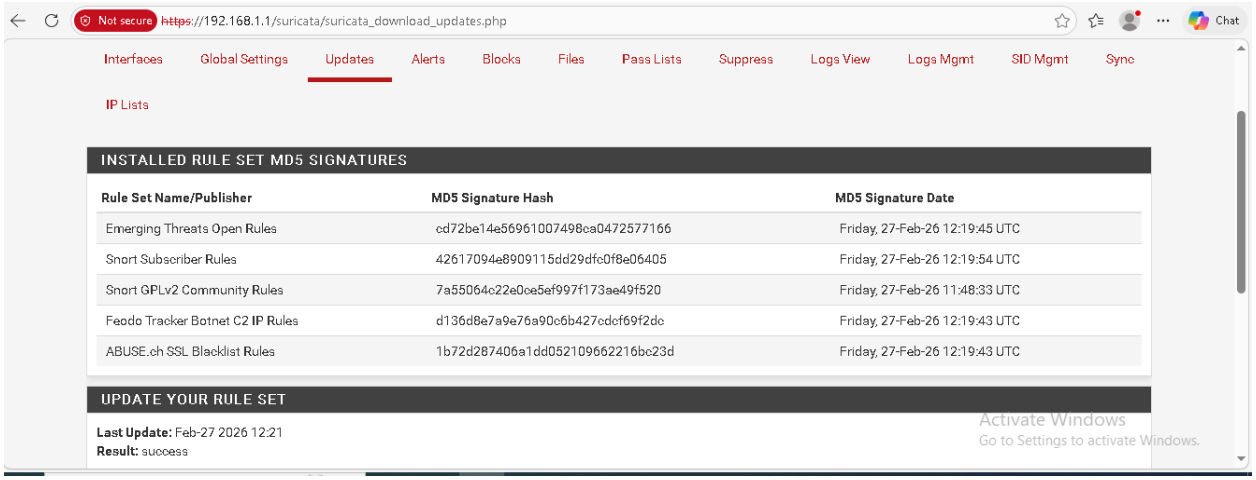


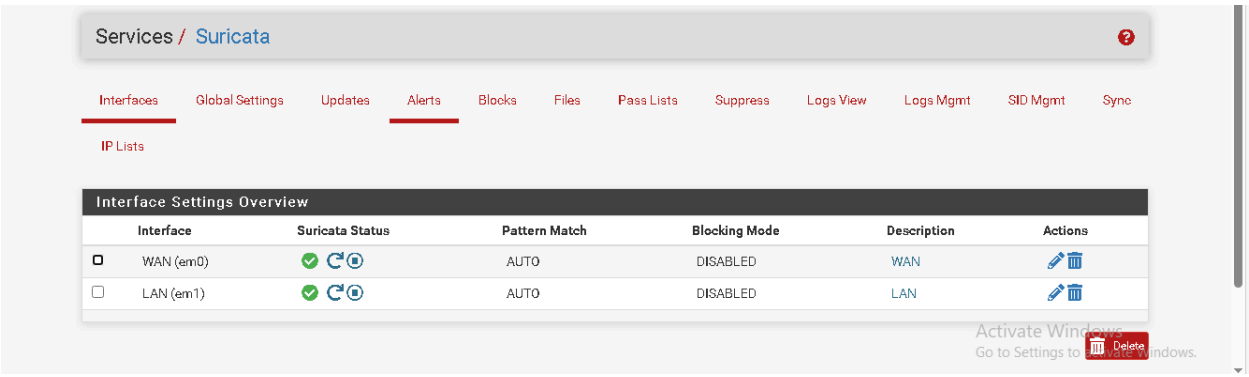
Description for Validating IDS Alerts in pfSense (Suricata): This screenshot from the pfSense web interface confirms that Suricata is actively inspecting traffic and generating alerts. The Alert Log View shows multiple instances of UDP traffic from the compromised host (192.168.1.100) triggering "SURICATA QUIC failed decrypt" alerts.



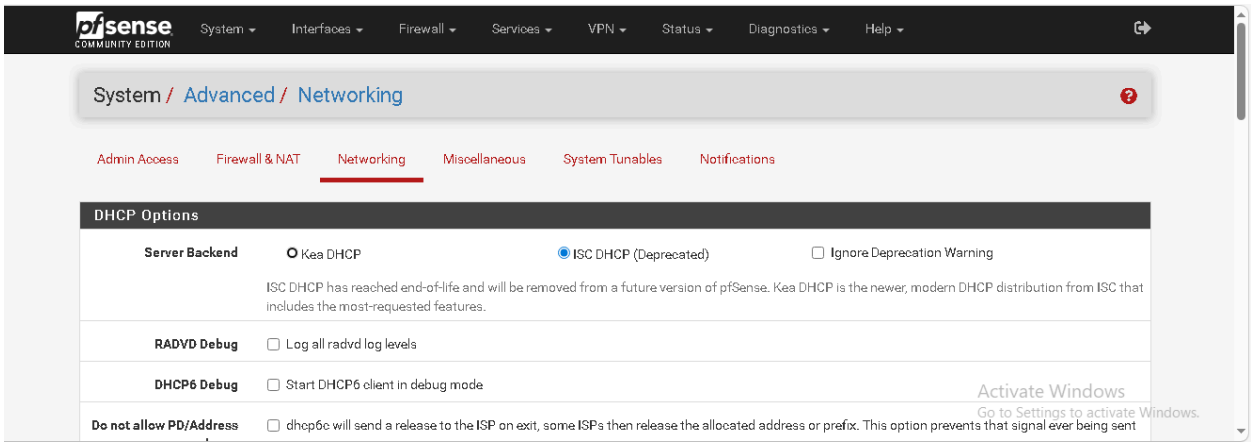
Make sure everything is updated ..

Description for Suricata Rule Set Verification: To maintain an effective defense posture, IDS signatures must be current. This screenshot shows the pfSense Suricata Updates tab, confirming that the Emerging Threats Open Rules, Snort Subscriber Rules, and various IP Blacklists have been successfully downloaded and applied.

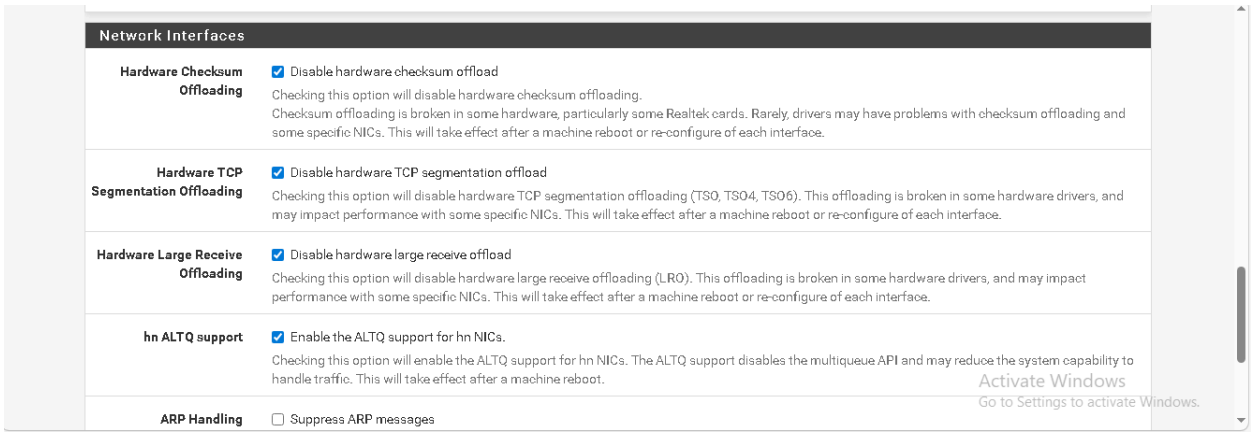




Description for Suricata Interface Configuration: This view of the Suricata Interface Settings Overview confirms that the IDS engine is actively bound to and inspecting both the WAN (em0) and LAN (em1) interfaces, ensuring complete visibility of ingress, egress, and lateral traffic.



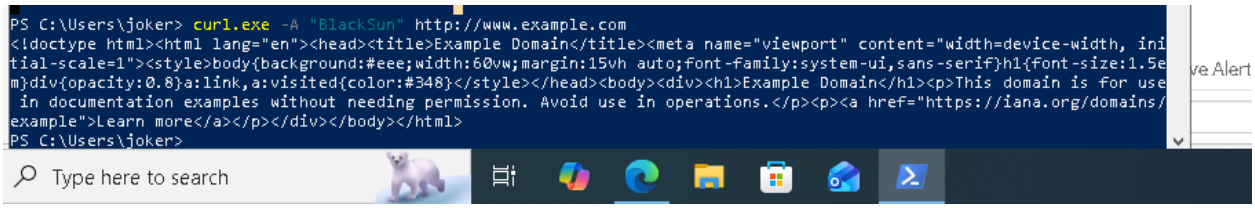
Description for Disabling Hardware Checksum Offloading: For an IDS like Suricata or Snort to inspect packets correctly, they must see the raw packet data. This pfSense configuration screenshot shows that Hardware Checksum Offloading has been intentionally disabled to prevent the network interface card (NIC) from altering packets before the IDS engine can analyze them.



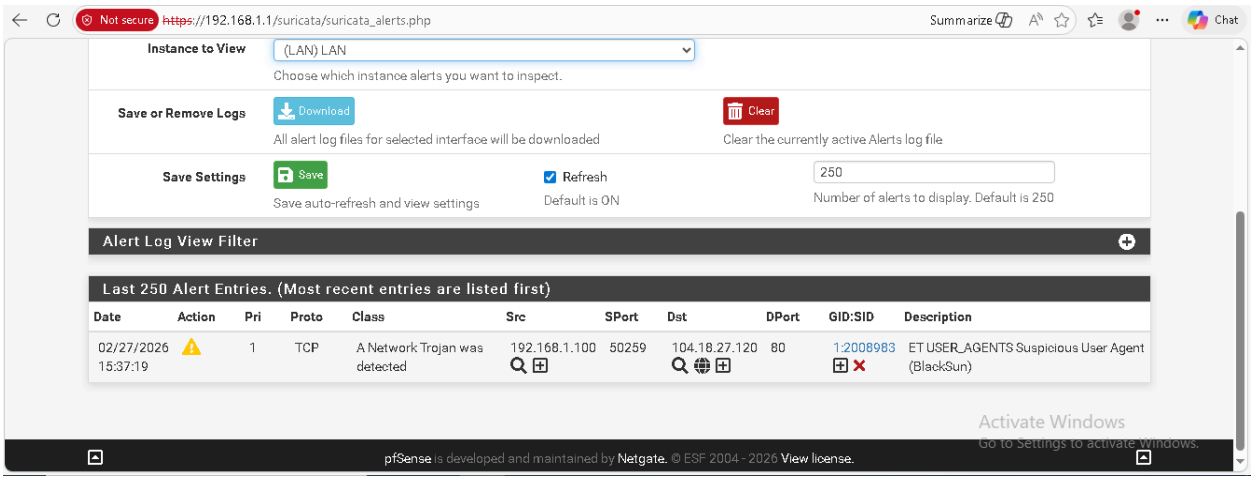
When you ran curl.exe -A "BlackSun" http://www.example.com, all that text that flooded your screen was not an error. That was just the raw HTML code of the example.com website being downloaded to your terminal. The command executed flawlessly.

Phase 5: Executing the "BlackSun" Malware Simulation

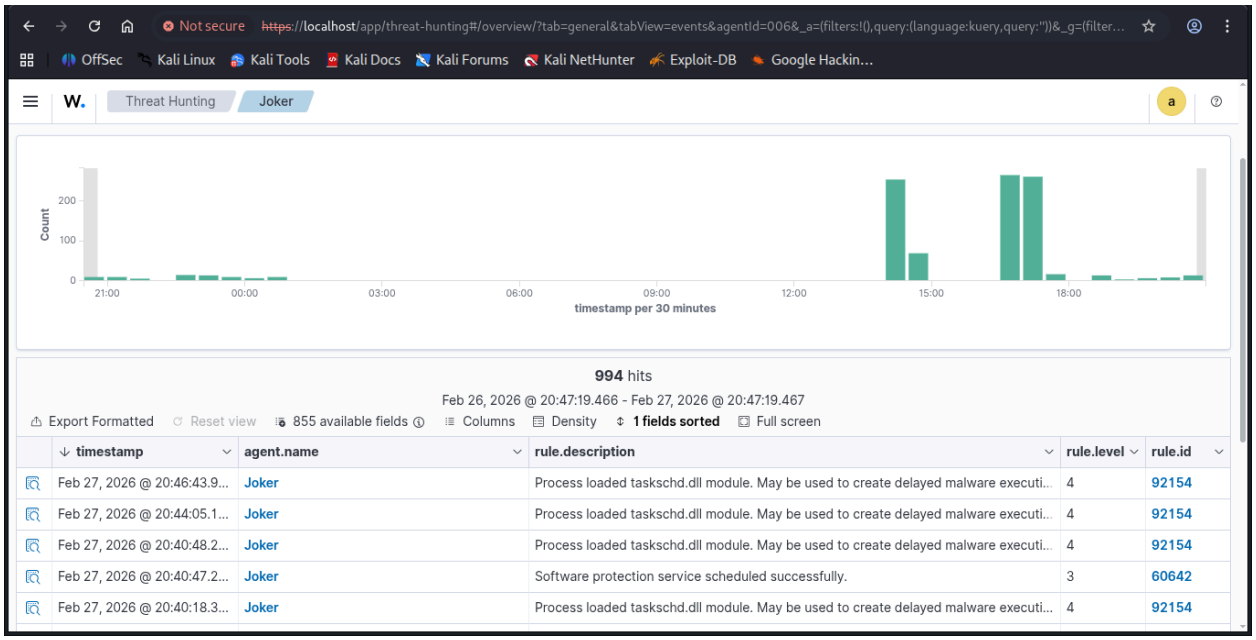
Description for Executing a Malicious User-Agent Payload: To simulate malware command and control (C2) beaconing, a curl command was executed with a custom, highly suspicious User-Agent string: -A "BlackSun" directed at http://www.example.com. The terminal output shows the successful retrieval of the raw HTML.



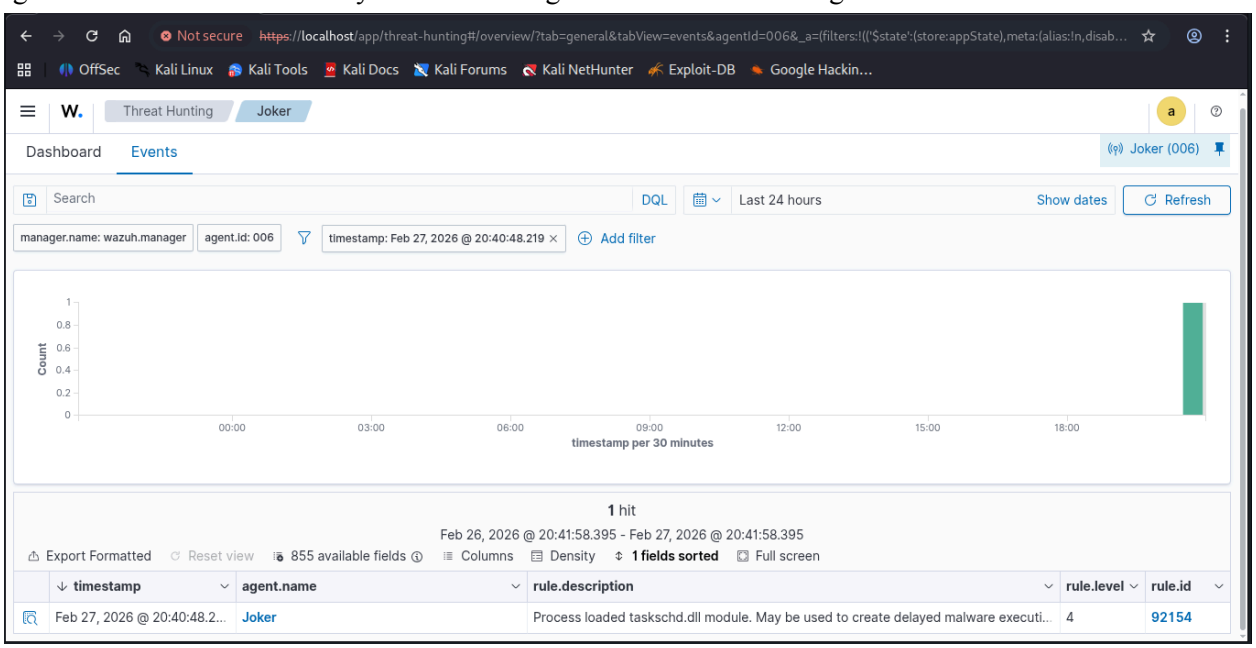
Description for NIDS Detection of the "BlackSun" Payload: The pfSense Suricata engine successfully intercepted the simulated C2 traffic. The Alert Log clearly displays that the rule ET USER_AGENTS Suspicious User Agent (BlackSun) fired against the traffic originating from 192.168.1.100, classifying it as a Network Trojan detection.



Description for Wazuh Dashboard Ingestion of Sysmon Events: This screenshot proves that the endpoint telemetry generated by Sysmon is being successfully forwarded to the centralized Wazuh SIEM. The Threat Hunting dashboard shows alerts for process executions involving taskschd.dll, triggering rule 92154 (May be used to create delayed malware execution).



Description for Validating Wazuh Rule and Agent Communication: Expanding on the ingested logs, this Wazuh Discover view shows the raw JSON parsing of the endpoint events. It confirms the Wazuh agent named "Joker" is actively communicating with the "wazuh.manager".



Phase 6: SIEM Integration and Centralized Logging

Description for Configuring pfSense Log Ingestion: To correlate network alerts with endpoint data, the Wazuh Manager must be configured to ingest logs from pfSense. This screenshot shows the ossec.conf file being modified using the Pluma text editor to add a <localfile> block pointing to /var/log/pfsense.log.

Configure Wazuh to Ingest pfSense Logs

On the Wazuh Manager, edit the configuration file:

```
<!-- Add to /var/ossec/etc/ossec.conf -->

<localfile>

  <log_format>syslog</log_format>

  <location>/var/log/pfsense.log</location>

</localfile>
```

ossec.conf (/var/ossec/etc) - Pluma

File Edit View Search Tools Documents Help

Open Save Undo

ossec.conf

```
  <localfile>
    <log_format>apache</log_format>
    <location>/var/log/apache2/error.log</location>
  </localfile>

  <localfile>
    <log_format>apache</log_format>
    <location>/var/log/apache2/access.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/pfsense.log</location>
  </localfile>

<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>192.168.137.0/24</allowed-ips>
</remote>

</ossec_config>
```

HTML Tab Width: 4 Ln 225, Col 15 INS

Create Custom Decoders for pfSense

pfSense logs need proper parsing to extract meaningful fields:

```
<!-- Custom decoder for pfSense -->
<decoder name="pfsense">
  <program_name>^pfsense</program_name>
</decoder>

<decoder name="pfsense-firewall">
  <parent>pfsense</parent>
  <prematch>filterlog:</prematch>
  <regex offset="after_prematch">

^(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\S+),(\d+),(\S+),(\S+),(\d+),(
\S+),(\d+),(\d+),(\S+),(\S+),(\d+),(\d+),(\d+)
  </regex>
  <order>
    timestamp,rule_number,sub_rule_number,anchor,tracker,interface,reason,action,direction,
    ip_version,tos,ecn,ttl,id,offset,flags,proto_id,proto,length,src_ip,dst_ip,src_port,
    dst_port,data_length,tcp_flags,sequence_number
  </order>
</decoder>
```

Description for Creating Custom Decoders for pfSense: Because pfSense logs use a specific structure, a custom Wazuh decoder (01-pfsense-decoder.xml) was authored. This regex-based decoder extracts critical fields such as source IP, destination IP, ports, and action from the raw pfSense filterlog outputs.

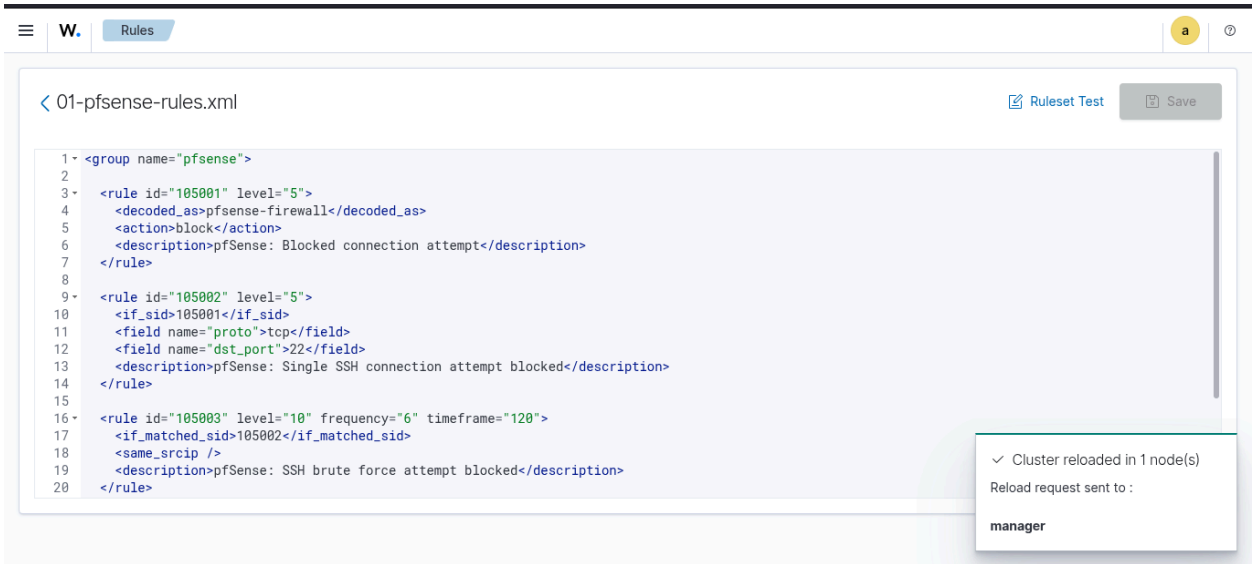
Description for Creating Custom Alert Rules:

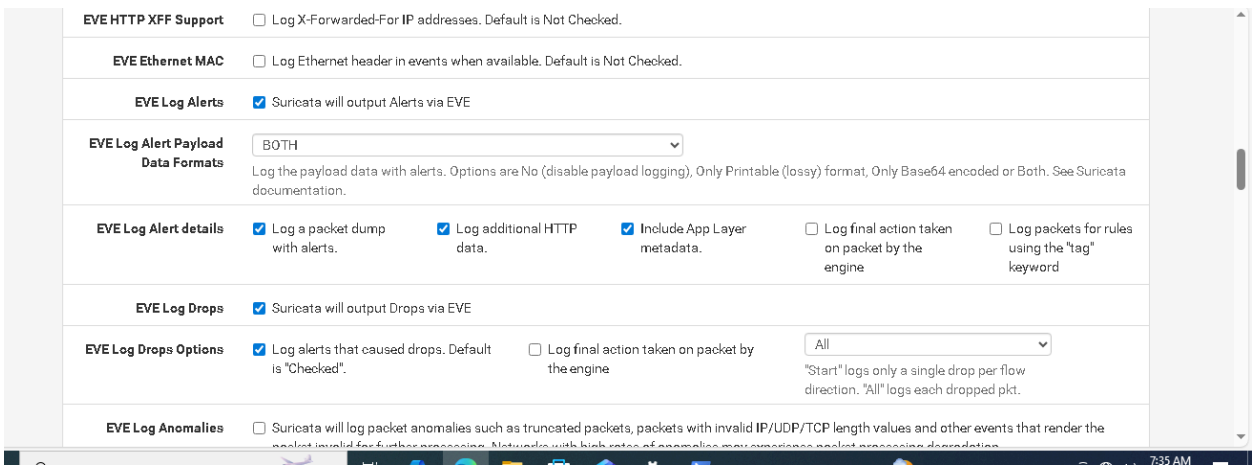
With the decoders extracting the data, custom Wazuh rules (01-pfsense-rules.xml) were created to trigger alerts based on specific logic. The screenshot shows rules designed to detect blocked connection attempts, SSH brute-force attacks, and lateral movement originating from internal IPs.

Create Custom Rules for pfSense

High-priority firewall events:

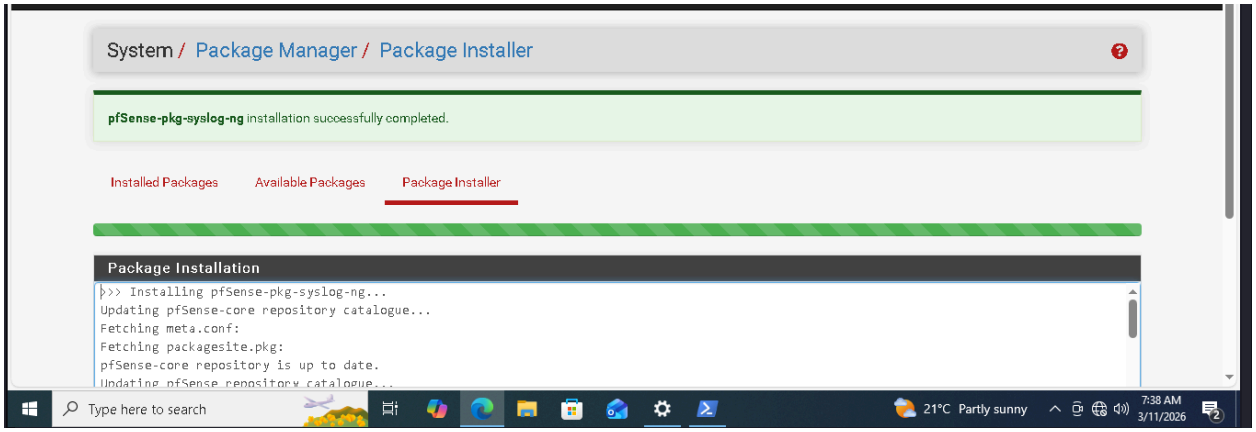
```
<group name="pfsense">

  <rule id="105001" level="5">
    <decoded_as>pfsense-firewall</decoded_as>
    <action>block</action>
    <description>pfSense: Blocked connection attempt</description>
  </rule>

  <rule id="105002" level="5">
    <if_sid>105001</if_sid>
    <field name="proto">tcp</field>
    <field name="dst_port">22</field>
    <description>pfSense: Single SSH connection attempt blocked</description>
  </rule>

  <rule id="105003" level="10" frequency="6" timeframe="120">
    <if_matched_sid>105002</if_matched_sid>
    <same_srcip />
    <description>pfSense: SSH brute force attempt blocked</description>
  </rule>

  <rule id="105004" level="10">
    <if_sid>105001</if_sid>
    <field name="src_ip" type="pcre2">^192\.168\.</field>
    <description>pfSense: Internal IP blocked (possible lateral movement)</description>
  </rule>

</group>
```





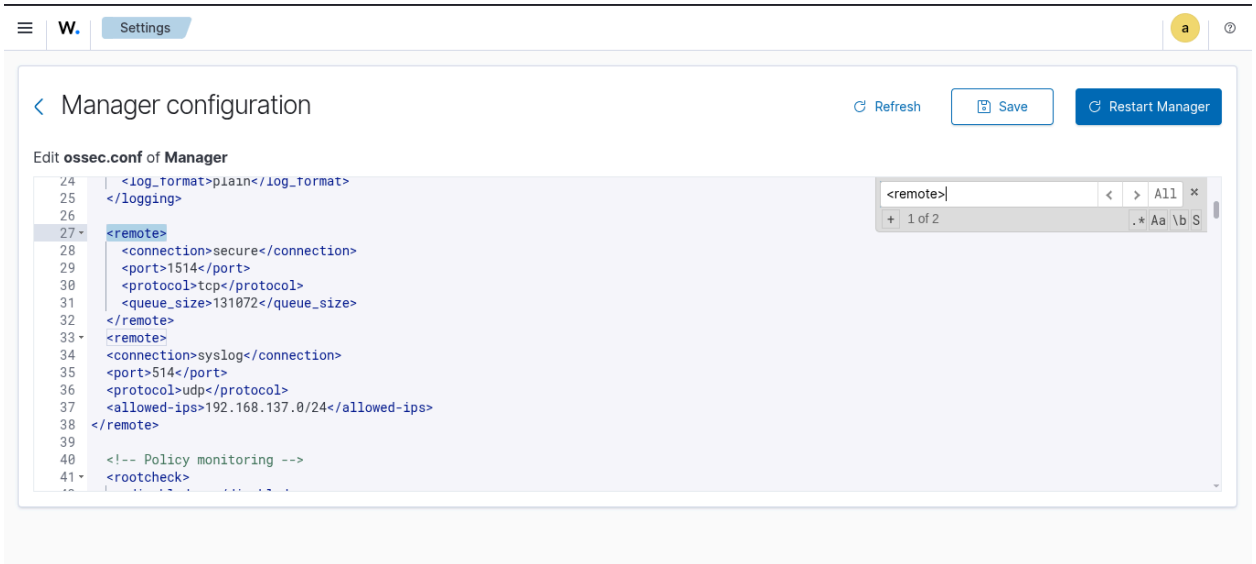
Description for Installing syslog-ng on pfSense: To securely and reliably forward Suricata alerts from the pfSense firewall to the Wazuh SIEM, the syslog-ng package was installed. The screenshot confirms the successful installation via the pfSense Package Installer.

Install syslog-ng in pfsense:



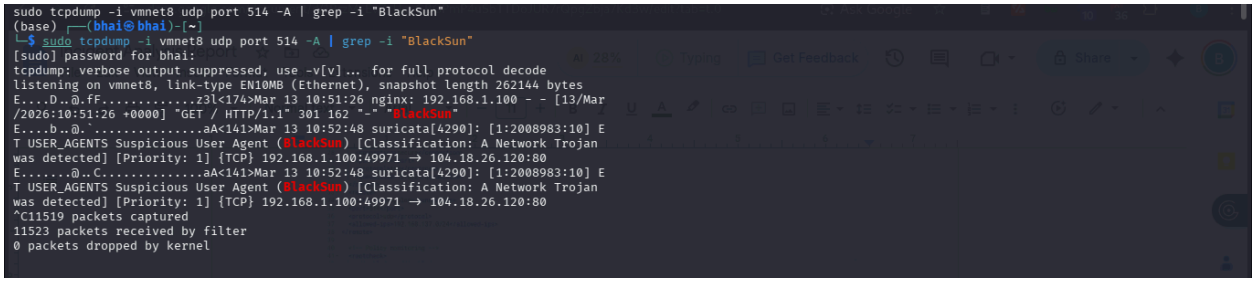
Make sure you have the remove tabs in the Server Management->Settings->edit Configuration

Description for Enabling Remote Syslog Reception in Wazuh: To accept the incoming syslog-ng stream from pfSense, the Wazuh Manager's ossec.conf was modified. This screenshot shows the <remote> block being configured to listen on UDP port 514 and explicitly allowing connections from the pfSense IP subnet (192.168.137.0/24).



Description for Packet Capture Verification (tcpdump): To troubleshoot and verify that the logs were actually traversing the network, a tcpdump was run on the Kali Linux host machine. The terminal output confirms that the raw syslog traffic containing the "BlackSun" User-Agent alert was successfully received on UDP port 514.

You can Check if the Blacksun is being sent to the udp open port 514 from the pfsense:



1. The Windows Agent Caught You (Sysmon)

The Wazuh agent on your Windows machine (named **Joker**, IP 192.168.1.100) successfully caught you executing the command.

1. It logged that someone used PowerShell to launch curl.exe.
2. It captured your exact command line: "C:\Windows\system32\curl.exe" -A BlackSun http://192.168.1.1 and http://www.example.com.
3. Sysmon even mapped it to a MITRE ATT&CK technique: **T1105 (Ingress Tool Transfer)**.

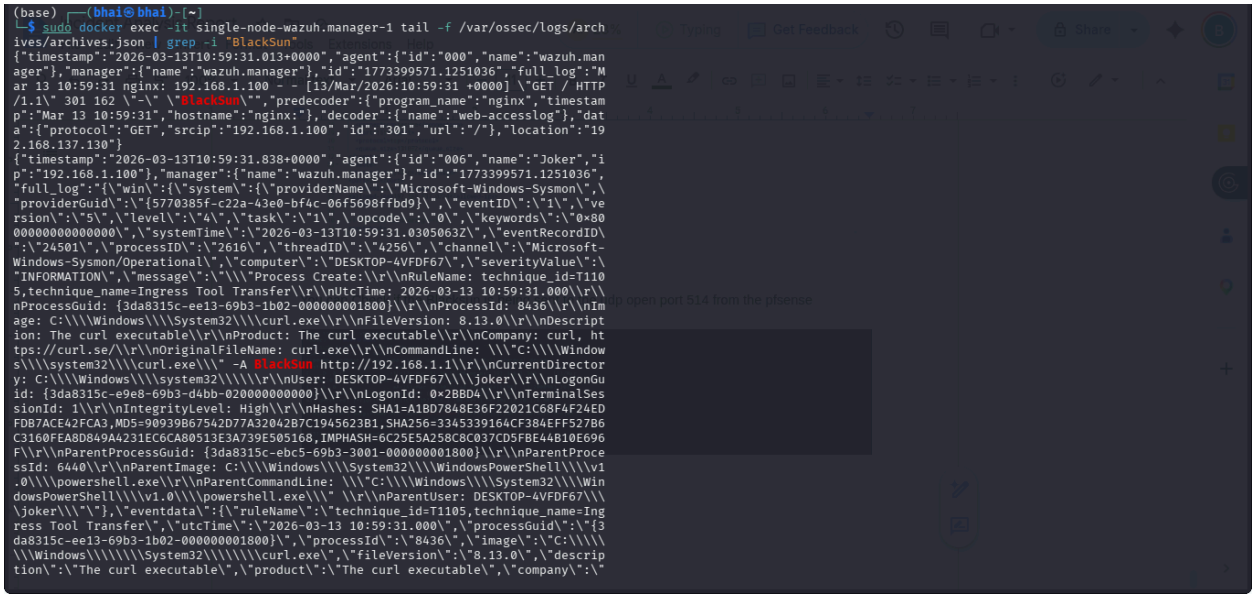
2. The Network Monitor Caught You (Suricata)

Wazuh also caught the traffic over the network. Suricata (an Intrusion Detection System) saw your web request flying across the network, noticed the user agent was "BlackSun", and flagged it as:

1. [Classification: A Network Trojan was detected] [Priority: 1]

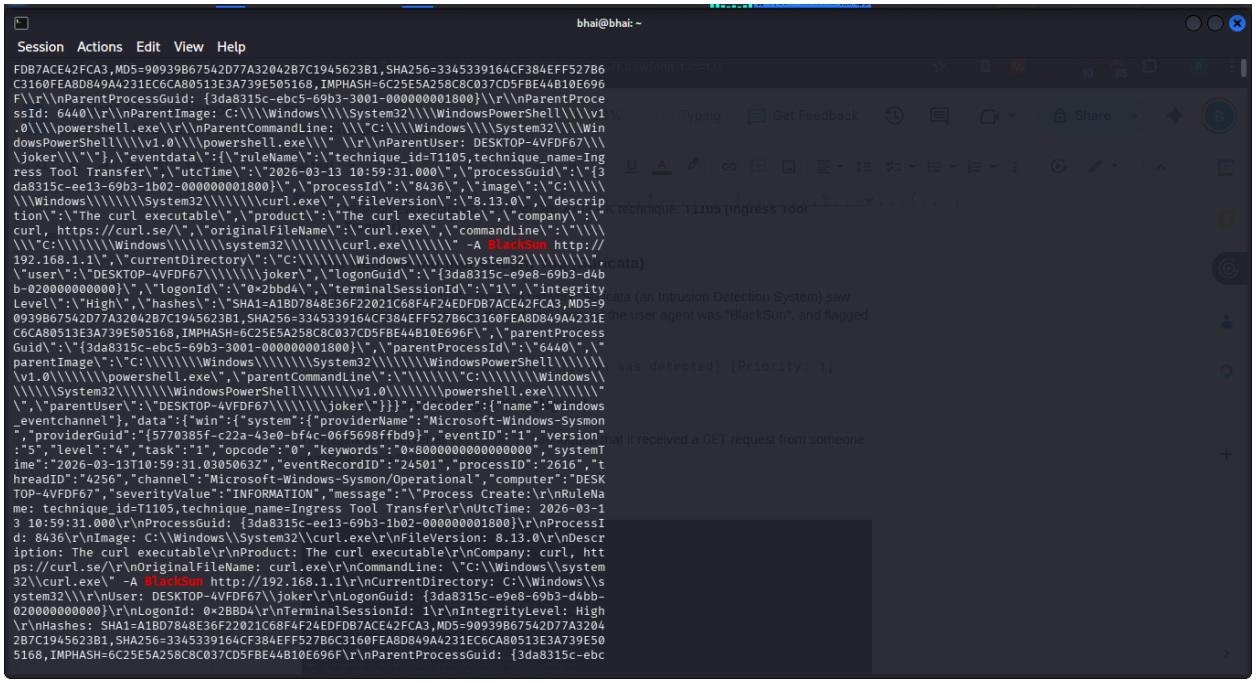
3. The Web Server Caught You (Nginx)

The Nginx web server at 192.168.1.100 logged that it received a GET request from someone using the "BlackSun" user agent.

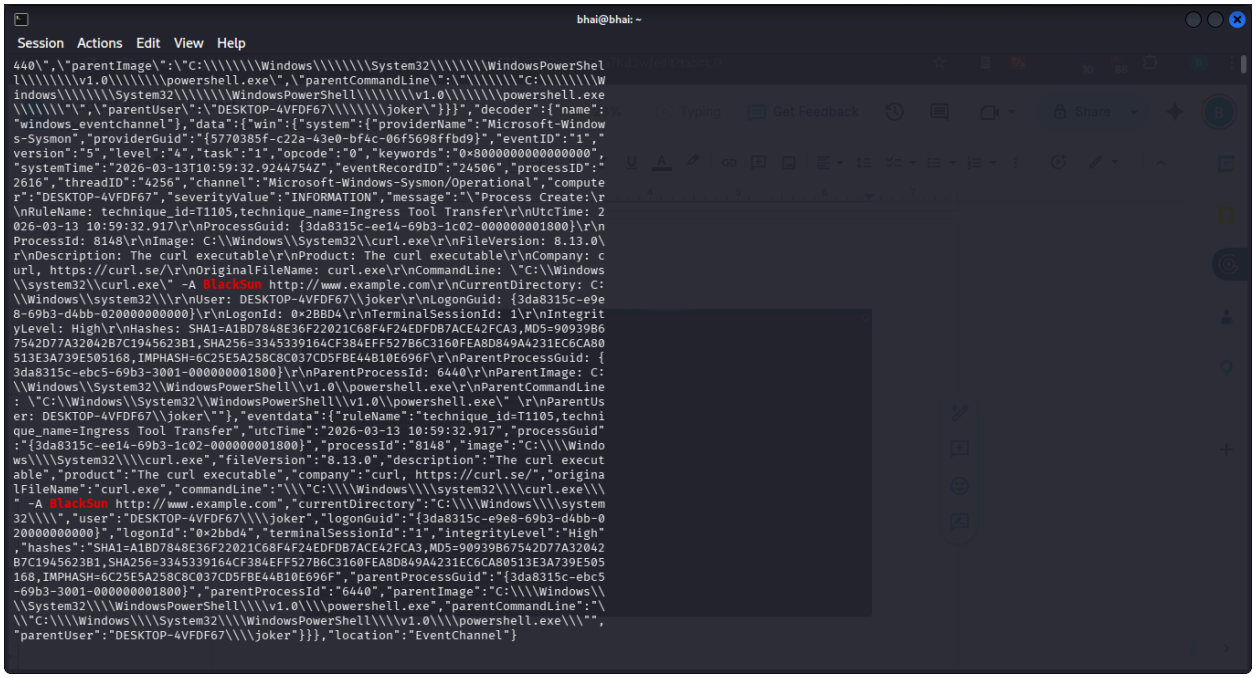


```
(base) ┌──(bhai㉿bhai)-[~]
└─$ sudo docker exec -it single-node-wazuh.manager-1 tail -f /var/ossec/logs/arch
ives/archives.json | grep -i "BlackSun"
{"timestamp":"2026-03-13T10:59:31.013+0000","agent":{"id":"000","name":"wazuh.man
ager"},"manager":{"name":"wazuh.manager"},"id":"1773399571.1251036","full_log":"M
ar 13 10:59:31 nginx: 192.168.1.100 - - [13/Mar/2026:10:59:31 +0000] \"GET / HTTP
/1.1\" 301 162 \"-\" \"BlackSun\"","predecoder":{"program_name":"nginx","timestam
p":"Mar 13 10:59:31","hostname":"nginx:"},"decoder":{"name":"web-accesslog"},"dat
a":{"protocol":"GET","srcip":"192.168.1.100","id":"301","url":"/"},"location":"19
2.168.137.130"}
{"timestamp":"2026-03-13T10:59:31.838+0000","agent":{"id":"006","name":"Joker","i
p":"192.168.1.100"},"manager":{"name":"wazuh.manager"},"id":"1773399571.1251036",
"full_log":"{\"win\":{\"system\":{\"providerName\":\"Microsoft-Windows-Sysmon\",\
"providerGuid\":\"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}\",\"eventID\":\"1\",\"ve
rsion\":\"5\",\"level\":\"4\",\"task\":\"1\",\"opcode\":\"0\",\"keywords\":\"0x80
00000000000000\",\"systemTime\":\"2026-03-13T10:59:31.0305063Z\",\"eventRecordID\
":\"24501\",\"processID\":\"2616\",\"threadID\":\"4256\",\"channel\":\"Microsoft-
Windows-Sysmon/Operational\",\"computer\":\"DESKTOP-4VFDF67\",\"severityValue\":\
"INFORMATION\",\"message\":\"\\\"Process Create:\\r\\nRuleName: technique_id=T110
5,technique_name=Ingress Tool Transfer\\r\\nUtcTime: 2026-03-13 10:59:31.000\\r\\
nProcessGuid: {3da8315c-ee13-69b3-1b02-000000001800}\\r\\nProcessId: 8436\\r\\nIm
age: C:\\\\Windows\\\\System32\\\\curl.exe\\r\\nFileVersion: 8.13.0\\r\\nDescript
ion: The curl executable\\r\\nProduct: The curl executable\\r\\nCompany: curl, ht
tps://curl.se/\\r\\nOriginalFileName: curl.exe\\r\\nCommandLine: \\\"C:\\\\Window
s\\\\system32\\\\curl.exe\\\" -A BlackSun http://192.168.1.1\\r\\nCurrentDirector
y: C:\\\\Windows\\\\system32\\\\\\r\\nUser: DESKTOP-4VFDF67\\\\joker\\r\\nLogonGu
id: {3da8315c-e9e8-69b3-d4bb-020000000000}\\r\\nLogonId: 0×2BBD4\\r\\nTerminalSes
sionId: 1\\r\\nIntegrityLevel: High\\r\\nHashes: SHA1=A1BD7848E36F22021C68F4F24ED
FDB7ACE42FCA3,MD5=90939B67542D77A32042B7C1945623B1,SHA256=3345339164CF384EFF527B6
C3160FEA8D849A4231EC6CA80513E3A739E505168,IMPHASH=6C25E5A258C8C037CD5FBE44B10E696
F\\r\\nParentProcessGuid: {3da8315c-ebc5-69b3-3001-000000001800}\\r\\nParentProce
ssId: 6440\\r\\nParentImage: C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1
.0\\\\powershell.exe\\r\\nParentCommandLine: \\\"C:\\\\Windows\\\\System32\\\\Win
dowsPowerShell\\\\v1.0\\\\powershell.exe\\\" \\r\\nParentUser: DESKTOP-4VFDF67\\\
\joker\\\"\"},\"eventdata\":{\"ruleName\":\"technique_id=T1105,technique_name=Ing
ress Tool Transfer\",\"utcTime\":\"2026-03-13 10:59:31.000\",\"processGuid\":\"{3
da8315c-ee13-69b3-1b02-000000001800}\",\"processId\":\"8436\",\"image\":\"C:\\\\\
\\\Windows\\\\\\\\System32\\\\\\\\curl.exe\",\"fileVersion\":\"8.13.0\",\"descrip
tion\":\"The curl executable\",\"product\":\"The curl executable\",\"company\":\"
```

Description for Raw Log Verification in Wazuh Archives: As final proof of ingestion, the /var/ossec/logs/archives/archives.json file was tailed on the Wazuh Manager. The output shows the raw, unparsed JSON log containing the entire process creation event from Sysmon (Event ID 1) detailing the exact PowerShell curl.exe command used in the BlackSun simulation.



```
bhai@bhai: ~
Session  Actions  Edit  View  Help
FDB7ACE42FCA3,MD5=90939B67542D77A32042B7C1945623B1,SHA256=3345339164CF384EFF527B6
C3160FEA8D849A4231EC6CA80513E3A739E505168,IMPHASH=6C25E5A258C8C037CD5FBE44B10E696
F\\r\\nParentProcessGuid: {3da8315c-ebc5-69b3-3001-000000001800}\\r\\nParentProce
ssId: 6440\\r\\nParentImage: C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1
.0\\\\powershell.exe\\r\\nParentCommandLine: \\\"C:\\\\Windows\\\\System32\\\\Win
dowsPowerShell\\\\v1.0\\\\powershell.exe\\\" \\r\\nParentUser: DESKTOP-4VFDF67\\\
\joker\\\"\"},\"eventdata\":{\"ruleName\":\"technique_id=T1105,technique_name=Ing
ress Tool Transfer\",\"utcTime\":\"2026-03-13 10:59:31.000\",\"processGuid\":\"{3
da8315c-ee13-69b3-1b02-000000001800}\",\"processId\":\"8436\",\"image\":\"C:\\\\\
\\\Windows\\\\\\\\System32\\\\\\\\curl.exe\",\"fileVersion\":\"8.13.0\",\"descrip
tion\":\"The curl executable\",\"product\":\"The curl executable\",\"company\":\"
curl, https://curl.se/\",\"originalFileName\":\"curl.exe\",\"commandLine\":\"\\\\
\\\"C:\\\\\\\\Windows\\\\\\\\system32\\\\\\\\curl.exe\\\\\\\" -A BlackSun http://
192.168.1.1\",\"currentDirectory\":\"C:\\\\\\\\Windows\\\\\\\\system32\\\\\\\\",
\"user\":\"DESKTOP-4VFDF67\\\\\\\\joker\",\"logonGuid\":\"{3da8315c-e9e8-69b3-d4b
b-020000000000}\",\"logonId\":\"0×2bbd4\",\"terminalSessionId\":\"1\",\"integrity
Level\":\"High\",\"hashes\":\"SHA1=A1BD7848E36F22021C68F4F24EDFDB7ACE42FCA3,MD5=9
0939B67542D77A32042B7C1945623B1,SHA256=3345339164CF384EFF527B6C3160FEA8D849A4231E
C6CA80513E3A739E505168,IMPHASH=6C25E5A258C8C037CD5FBE44B10E696F\",\"parentProcess
Guid\":\"{3da8315c-ebc5-69b3-3001-000000001800}\",\"parentProcessId\":\"6440\",\"
parentImage\":\"C:\\\\\\\\Windows\\\\\\\\System32\\\\\\\\WindowsPowerShell\\\\\\\\
\v1.0\\\\\\\\powershell.exe\",\"parentCommandLine\":\"\\\\\\\"C:\\\\\\\\Windows\\\
\\\\System32\\\\\\\\WindowsPowerShell\\\\\\\\v1.0\\\\\\\\powershell.exe\\\\\\\"
\",\"parentUser\":\"DESKTOP-4VFDF67\\\\\\\\joker\"}}}","decoder":{"name":"windows
_eventchannel"},"data":{"win":{"system":{"providerName":"Microsoft-Windows-Sysmon
","providerGuid":"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}","eventID":"1","version"
:"5","level":"4","task":"1","opcode":"0","keywords":"0×8000000000000000","systemT
ime":"2026-03-13T10:59:31.0305063Z","eventRecordID":"24501","processID":"2616","t
hreadID":"4256","channel":"Microsoft-Windows-Sysmon/Operational","computer":"DESK
TOP-4VFDF67","severityValue":"INFORMATION","message":"\"Process Create:\r\nRuleNa
me: technique_id=T1105,technique_name=Ingress Tool Transfer\r\nUtcTime: 2026-03-1
3 10:59:31.000\r\nProcessGuid: {3da8315c-ee13-69b3-1b02-000000001800}\r\nProcessI
d: 8436\r\nImage: C:\\Windows\\System32\\curl.exe\r\nFileVersion: 8.13.0\r\nDescr
iption: The curl executable\r\nProduct: The curl executable\r\nCompany: curl, htt
ps://curl.se/\r\nOriginalFileName: curl.exe\r\nCommandLine: \"C:\\Windows\\system
32\\curl.exe\" -A BlackSun http://192.168.1.1\r\nCurrentDirectory: C:\\Windows\\s
ystem32\\\r\nUser: DESKTOP-4VFDF67\\joker\r\nLogonGuid: {3da8315c-e9e8-69b3-d4bb-
020000000000}\r\nLogonId: 0×2BBD4\r\nTerminalSessionId: 1\r\nIntegrityLevel: High
\r\nHashes: SHA1=A1BD7848E36F22021C68F4F24EDFDB7ACE42FCA3,MD5=90939B67542D77A3204
2B7C1945623B1,SHA256=3345339164CF384EFF527B6C3160FEA8D849A4231EC6CA80513E3A739E50
5168,IMPHASH=6C25E5A258C8C037CD5FBE44B10E696F\r\nParentProcessGuid: {3da8315c-ebc
```


bhai@bhai: ~
Session Actions Edit View Help
440\",\"parentImage\":\"C:\\\\\\\\Windows\\\\\\\\System32\\\\\\\\WindowsPowerShel
l\\\\\\\\v1.0\\\\\\\\powershell.exe\",\"parentCommandLine\":\"\\\\\\\"C:\\\\\\\\W
indows\\\\\\\\System32\\\\\\\\WindowsPowerShell\\\\\\\\v1.0\\\\\\\\powershell.exe
\\\\\\\"\",\"parentUser\":\"DESKTOP-4VFDF67\\\\\\\\joker\"}}}\",\"decoder":{"name":
"windows_eventchannel"},"data":{"win":{"system":{"providerName":"Microsoft-Window
s-Sysmon","providerGuid":"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}","eventID":"1","
version":"5","level":"4","task":"1","opcode":"0","keywords":"0x8000000000000000",
"systemTime":"2026-03-13T10:59:32.9244754Z","eventRecordID":"24506","processID":"
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r":"DESKTOP-4VFDF67","severityValue":"INFORMATION","message":"\"Process Create:\r
\nRuleName: technique_id=T1105,technique_name=Ingress Tool Transfer\r\nUtcTime: 2
026-03-13 10:59:32.917\r\nProcessGuid: {3da8315c-ee14-69b3-1c02-000000001800}\r\n
ProcessId: 8148\r\nImage: C:\\Windows\\System32\\curl.exe\r\nFileVersion: 8.13.0\
r\nDescription: The curl executable\r\nProduct: The curl executable\r\nCompany: c
url, https://curl.se/\r\nOriginalFileName: curl.exe\r\nCommandLine: \"C:\\Windows
\\system32\\curl.exe\" -A BlackSun http://www.example.com\r\nCurrentDirectory: C:
\\Windows\\system32\\\r\nUser: DESKTOP-4VFDF67\\joker\r\nLogonGuid: {3da8315c-e9e
8-69b3-d4bb-020000000000}\r\nLogonId: 0x2BBD4\r\nTerminalSessionId: 1\r\nIntegrit
yLevel: High\r\nHashes: SHA1=A1BD7848E36F22021C68F4F24EDFDB7ACE42FCA3,MD5=90939B6
7542D77A32042B7C1945623B1,SHA256=3345339164CF384EFF527B6C3160FEA8D849A4231EC6CA80
513E3A739E505168,IMPHASH=6C25E5A258C8C037CD5FBE44B10E696F\r\nParentProcessGuid: {
3da8315c-ebc5-69b3-3001-000000001800}\r\nParentProcessId: 6440\r\nParentImage: C:
\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\r\nParentCommandLine
: \"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" \r\nParentUs
er: DESKTOP-4VFDF67\\joker\""},"eventdata":{"ruleName":"technique_id=T1105,techni
que_name=Ingress Tool Transfer","utcTime":"2026-03-13 10:59:32.917","processGuid"
:"{3da8315c-ee14-69b3-1c02-000000001800}","processId":"8148","image":"C:\\\\Windo
ws\\\\System32\\\\curl.exe","fileVersion":"8.13.0","description":"The curl execut
able","product":"The curl executable","company":"curl, https://curl.se/","origina
lFileName":"curl.exe","commandLine":"\\\"C:\\\\Windows\\\\system32\\\\curl.exe\\\
" -A BlackSun http://www.example.com","currentDirectory":"C:\\\\Windows\\\\system
32\\\\","user":"DESKTOP-4VFDF67\\\\joker","logonGuid":"{3da8315c-e9e8-69b3-d4bb-0
20000000000}","logonId":"0x2bbd4","terminalSessionId":"1","integrityLevel":"High"
,"hashes":"SHA1=A1BD7848E36F22021C68F4F24EDFDB7ACE42FCA3,MD5=90939B67542D77A32042
B7C1945623B1,SHA256=3345339164CF384EFF527B6C3160FEA8D849A4231EC6CA80513E3A739E505
168,IMPHASH=6C25E5A258C8C037CD5FBE44B10E696F","parentProcessGuid":"{3da8315c-ebc5
-69b3-3001-000000001800}","parentProcessId":"6440","parentImage":"C:\\\\Windows\\
\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe","parentCommandLine":"\
\\"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\"",
"parentUser":"DESKTOP-4VFDF67\\\\joker"}}},"location":"EventChannel"}

Phase 7: Incident Response Plan - Malware Containment Simulation

Description for Generating the EICAR Standard Malware File: To simulate a live malware drop and test the endpoint's active response capabilities, a PowerShell command (Set-Content) was used to write the EICAR standard anti-virus test file string to the Desktop. Immediately upon file creation, Microsoft Defender Antivirus intercepted the payload, triggering a "Threats found" notification.

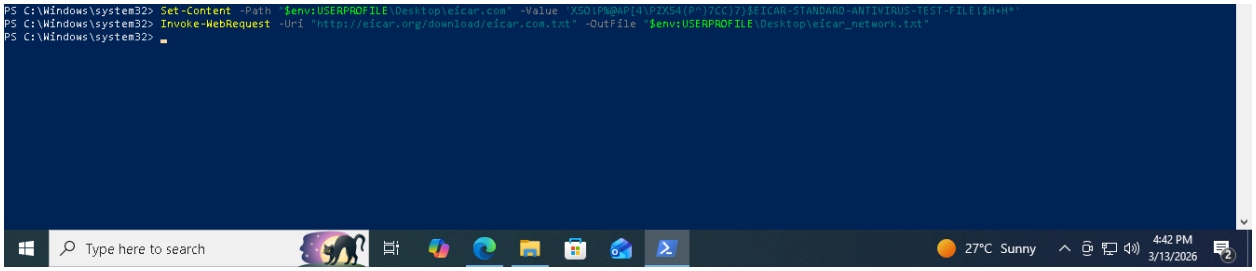
The EICAR Standard Malware File

The cybersecurity industry created a harmless string of text specifically designed to trigger Antivirus and EDR (Endpoint Detection and Response) systems. It is completely safe, but every security tool in the world is hardcoded to flag it as critical malware.

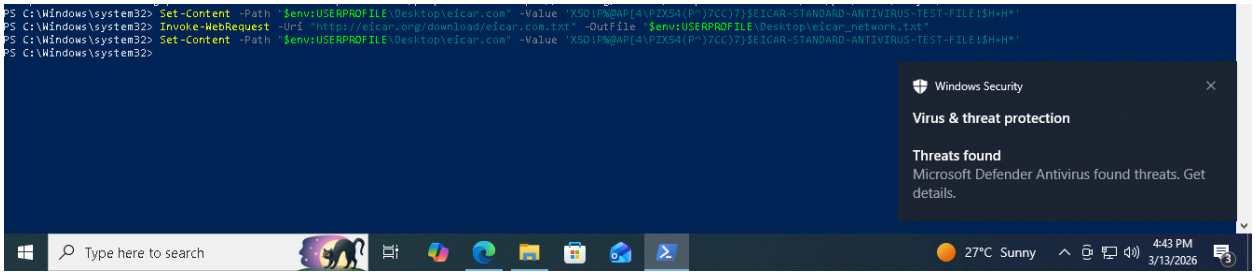
Open PowerShell on your target machine and run this command to create the fake malware file on your desktop:

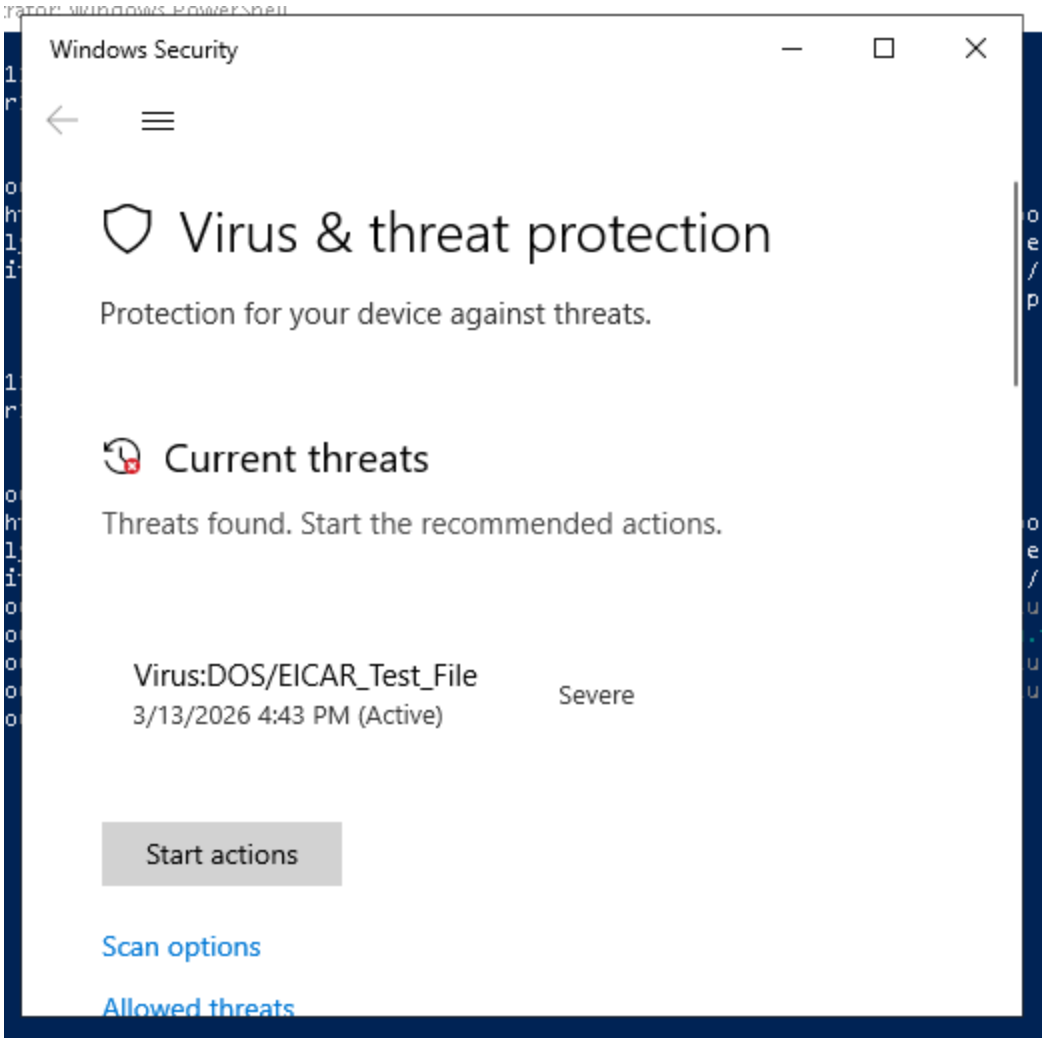
```
Set-Content -Path "$env:USERPROFILE\Desktop\eicar.com" -Value
'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*'
```

(Wazuh's File Integrity Monitoring or Windows Defender will instantly catch this, quarantine it, and send a screaming alert to your manager).



Description for Windows Defender Quarantine Verification: Opening the Windows Security dashboard reveals the specifics of the intervention. The system successfully identified the dropped artifact as Virus:DOS/EICAR_Test_File and assigned it a "Severe" threat level, effectively quarantining the simulated malware.





Description for SIEM Visibility of Malware Execution: Simultaneous to the local Windows Defender block, the event was logged by Sysmon and forwarded to the Wazuh Manager. This final screenshot displays the Wazuh Threat Hunting dashboard filtering for "eicar", showing multiple Level 6 alerts for "Executable file created by powershell" (Rule ID 92203), proving complete visibility into the incident response lifecycle.

Threat Hunting / Joker

Recently viewed

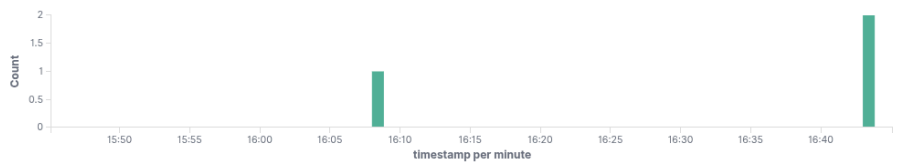
Home
Overview

Explore
Discover
Dashboards
Visualize
Reporting
Alerting
Anomaly Detection
Maps
Notifications

Endpoint security
Configuration Assessment

eicar | DQL | Refresh

manager.name: wazuh.manager | agent.id: 006 | Add filter



3 hits

Mar 13, 2026 @ 15:45:08.573 - Mar 13, 2026 @ 16:45:08.574

Export Formatted | Reset view | 856 available fields | Columns | Density | 1 fields sorted | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 13, 2026 @ 16:43:57.9...	Joker	Executable file created by powershell...	6	92203
Mar 13, 2026 @ 16:43:00.3...	Joker	Executable file created by powershell...	6	92203
Mar 13, 2026 @ 16:08:57.6...	Joker	Executable file created by powershell...	6	92203

OffSec | Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hackin...

Discover / Context of #WGoC55wBCR7vEjLdwlou

Recently viewed

Home
Overview

Explore
Discover
Dashboards
Visualize
Reporting
Alerting
Anomaly Detection
Maps
Notifications

Endpoint security
Configuration Assessment

Add filter

Load 5 newer documents

Only 1 documents newer than the anchor could be found.

Time	_source
Mar 13, 2026 @ 16:46:17.295	input.type: log agent.ip: 192.168.1.100 agent.name: Joker agent.id: 006 manager.name: wazuh.manager data.win.eventdata.param3: demand start data.win.eventdata.param4: BITS data.win.eventdata.param1: Background Intelligent Transfer Service data.win.eventdata.param2: auto start data.win.system.eventID: 7040 data.win.system.eventSourceName: Service Control Manager data.win.system.keywords: 0x8080000000000000 data.win.system.providerGuid: {555908d1-a6d7-4695-8e1e
Mar 13, 2026 @ 16:43:57.907	input.type: log agent.ip: 192.168.1.100 agent.name: Joker agent.id: 006 manager.name: wazuh.manager data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.processGuid: {3da8315c-ebc5-69b3-3001-000000001800} data.win.eventdata.processId: 6440 data.win.eventdata.utcTime: 2026-03-13 11:43:57.830 data.win.eventdata.targetFilename: C:\\Users\\joker\\Desktop\\eicar.com
Mar 13, 2026 @ 16:43:16.370	input.type: log agent.ip: 192.168.1.100 agent.name: Joker agent.id: 006 manager.name: wazuh.manager data.win.eventdata.param3: auto start data.win.eventdata.param4: BITS data.win.eventdata.param1: Background Intelligent Transfer Service data.win.eventdata.param2: demand start data.win.system.eventID: 7040 data.win.system.eventSourceName: Service Control Manager

Table JSON

@timestamp	Mar 13, 2026 @ 16:43:57.907
_index	wazuh-alerts-4.x-2026.03.13
agent.id	006
agent.ip	192.168.1.100
agent.name	Joker
data.win.eventdata.creationUtcTime	2026-03-13 11:43:57.830
data.win.eventdata.image	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe
data.win.eventdata.processGuid	{3da8315c-ebc5-69b3-3001-000000001800}
data.win.eventdata.processId	6440
data.win.eventdata.targetFilename	C:\\Users\\joker\\Desktop\\eicar.com
data.win.eventdata.user	DESKTOP-4VFDF67\\joker
data.win.eventdata.utcTime	2026-03-13 11:43:57.830
data.win.system.channel	Microsoft-Windows-Sysmon/Operational
data.win.system.computer	DESKTOP-4VFDF67
data.win.system.eventID	11
data.win.system.eventRecordID	25615

t	data.win.eventdata.user	DESKTOP-4VFDF67\\joker
t	data.win.eventdata.utcTime	2026-03-13 11:43:57.830
t	data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t	data.win.system.computer	DESKTOP-4VFDF67
t	data.win.system.eventID	11
t	data.win.system.eventRecordID	25615
t	data.win.system.keywords	0x8000000000000000
t	data.win.system.level	4
t	data.win.system.message	"File created: RuleName: - UtcTime: 2026-03-13 11:43:57.830 ProcessGuid: {3da8315c-ebc5-69b3-3001-000000001800} ProcessId: 6440 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe TargetFilename: C:\Users\joker\Desktop\eicar.com CreationUtcTime: 2026-03-13 11:43:57.830 User: DESKTOP-4VFDF67\joker"
t	data.win.system.opcode	0
t	data.win.system.processID	2616
t	data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t	data.win.system.providerName	Microsoft-Windows-Sysmon

t	data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t	data.win.system.providerName	Microsoft-Windows-Sysmon
t	data.win.system.severityValue	INFORMATION
t	data.win.system.systemTime	2026-03-13T11:43:57.8474409Z
t	data.win.system.task	11
t	data.win.system.threadID	4256
t	data.win.system.version	2
t	decoder.name	windows_eventchannel
t	id	1773402237.1460049
t	input.type	log
t	location	EventChannel
t	manager.name	wazuh.manager
t	rule.description	Executable file created by powershell: C:\\Users\\joker\\Desktop\\eicar.com
#	rule.firedtimes	3
t	rule.groups	sysmon, sysmon_eid11_detections, windows
t	rule.id	92203
#	rule.level	6
◕	rule.mail	false
t	rule.mitre.id	T1105

data.win.system.threadID	4256
data.win.system.version	2
decoder.name	windows_eventchannel
id	1773402237.1460049
input.type	log
location	EventChannel
manager.name	wazuh.manager
rule.description	Executable file created by powershell: C:\\Users\\joker\\Desktop\\eicar.com
rule.firedtimes	3
rule.groups	sysmon, sysmon_eid11_detections, windows
rule.id	92203
rule.level	6
rule.mail	false
rule.mitre.id	T1105
rule.mitre.tactic	Command and Control
rule.mitre.technique	Ingress Tool Transfer
timestamp	Mar 13, 2026 @ 16:43:57.907

1. **IOCs (Indicators of Compromise):** The static evidence left behind *after* the attack (the fingerprints).
2. **IOAs (Indicators of Attack):** The malicious behaviors and techniques happening *during* the attack (the act of breaking in).

Based entirely on the raw Sysmon and Suricata logs you just generated from your Joker machine, here are your exact indicators:

Indicators of Compromise (IOCs) - The Fingerprints

These are the specific artifacts you can feed into threat intelligence feeds, firewalls, or Wazuh CDB lists to instantly block future attacks.

1. File Artifacts:

1. **Malicious File Path:** C:\Users\joker\Desktop\eicar.com
2. **Malicious File String/Signature:** X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*

2. Network Artifacts:

1. **Suspicious User-Agent String:** BlackSun
2. **Suspicious Destination IP:** 104.18.26.120 (The IP Suricata flagged when your curl command reached out to the internet).
3. **Suspicious URL:** http://eicar.org/download/eicar.com.txt

Indicators of Attack (IOAs) - The Behaviors

These are the behaviors you want to write Wazuh rules for (like the MITRE ATT&CK mappings you saw in your logs). If you detect the *behavior*, you catch the hacker even if they change their IP address, file name, or malware hash.

1. Abnormal Process Lineage (Parent-Child Anomalies):

1. powershell.exe spawning curl.exe as a child process. (System administrators rarely do this, but hackers do it constantly to download payloads).
2. *Mapped in your logs to:* MITRE T1105 (Ingress Tool Transfer).

2. Suspicious File Creation Activity:

1. A command-line tool (powershell.exe) directly creating an executable file (.com or .exe) in a user-writable directory (\Desktop\).
2. *Mapped in your logs to:* Rule 92203.

3. Evasion & Reconnaissance Behavior:

1. Using the -A flag in curl.exe to intentionally spoof or alter a User-Agent header to bypass basic web filtering.
2. Clearing Windows Event Logs via wevtutil cl Setup (The Anti-Forensics test we discussed earlier).